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Measuring Risk Information

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ABSTRACT

We develop a measure of how information events impact investors' expectations of risk. The measure is broadly applicable and simple to implement. We derive it from an option-pricing model, where investors anticipate an announcement that simultaneously conveys information on the announcer's expected future cash flows and risk profile. We empirically implement the measure using firms' earnings announcements, showing that it closely aligns with our model's predictions and offers strong forecasting power for firms' risk profiles, costs of capital, and future investments. We further highlight pitfalls of using simple changes in option-implied volatilities to study information gleaned from earnings announcements. Finally, we apply our measure to study disclosure regulation, the efficacy of text-based proxies, and market-wide events, which we use to illustrate our measure's uses, and illuminate its potential limitations.

JEL codes: G10, G11, G12, G14, M40, M41

Keywords: risk uncertainty; earnings announcements; implied volatility; cost of capital; risk disclosure

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1. Introduction

Information is frequently impounded into stock prices in the form of discrete, anticipated events, such as earnings announcements, press releases, investor conferences, SEC filings, and policy announcements. Research on the impact of these events typically focuses on the information they contain regarding expected future cash flows. Yet, in many cases, these events simultaneously provide information that changes investors' expectations of risk, which we refer to as risk information. For example, investors may learn about the scope of firms' risky investments, their financial liquidity, exposures to risk factors, and/or changes in their operational strategy.

Our study is motivated by the importance of quantifying the risk information contained in these events. Expectations of risk play a central role in most research settings studied by financial economists. For example, the severity and nature of risk influences decisions over debt contracting, compensation design, information acquisition, and portfolio choice. Therefore, characterizing and quantifying risk information gleaned from information events is an important step toward understanding how and why interactions among economic agents evolve over time. We derive a measure of an anticipated event's risk information content, implement it empirically, and run a battery of tests that yield support for its efficacy.

As an example of how information events can provide information on risk, consider Facebook's earnings announcement in July 2018. This announcement commanded national headlines in part because it triggered a single-day stock price drop of 19% when providing information on risk in addition to information on cash flows.[1] Despite beating analysts' earnings forecasts, Facebook warned of heightened risks associated with privacy controls, concerns over user growth, and the fallout from the Cambridge Analytica scandal.[2]

Studying risk information conveyed by events like Facebook's announcement poses a challenge: How can researchers calibrate the risk information when the events also provide information on the level of future cash flows? Standard measures of the information content of an announcement based on stock prices cannot be used to quantify risk information for at least two reasons. First, under prevailing asset-pricing theory, expected stock returns are independent of firm-specific risks. Second, stock returns (as well as bond returns) are jointly and simultaneously influenced by information on expected cash flows and risk-factor exposures, which precludes

[1] See for example, Castillo [2018], Price [2018], and Phillips [2018].

[2] Earnings announcements can also reduce investors' expectations over risk. Take, for example, Raytheon's earnings announcement in February 2019, in which the company noted greater stability in their projected sales to the U.S government and an increase in capital returned to investors via dividends and buybacks. Our approach suggests that this announcement lowered investors' expectations of risk equivalent in magnitude to $\sim 25\%$ of the diffusive variance expected over the month following the announcement.

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using stock returns to distinguish these two types of information. To measure risk information, we turn our attention to option markets, which enable researchers to directly observe investors' expectations of the firm's future return variance.

We begin with an option-pricing model in which investors observe an anticipated information event that contains information on both the mean and variance of the firm's future payoffs. The model's central output is our measure of risk information, denoted as $RiskInfo_t$, which captures how the event impacts investors' expectations of the future return variance over the t days following its release. Following the "event-study" approach, $RiskIn\,fo_t$ measures the event's impact on investor beliefs by focusing on changes in market prices in a short window around the announcement.[3] Specifically, $RiskIn\,fo_t$ is based on the change in the return variance implied by options that mature t days after the announcement. The link between this change and risk information is intuitive: Because the option-implied variance captures investors' beliefs regarding the future return variance, its change around the event should capture how the event affected investors' beliefs about this variance.

However, our model shows that the change in the implied variance would exhibit a systematic downward bias if directly used as a metric of risk information. The reason is that, in expectation, option-implied variances predictably drop after the release of anticipated news events, regardless of whether these events contain risk information (Patell and Wolfson [1979], Dubinsky et al. [2019]). The intuition is that, because these events move prices, investors expect them to generate heightened return volatility, which causes the option-implied variance to rise prior to their release. After these events pass, volatility is no longer elevated, and thus the option-implied variance drops to its typical level, on average.

Our model shows that, to measure risk information, one simply needs to adjust the change in the implied variance on the event date for this expected decline. Moreover, for any given event, this expected decline can be measured and adjusted for using the difference in the implied variances from short- and long-dated options just prior to its release. Intuitively, the magnitude of the expected decline depends upon the amount of return variation investors expect the event to create. Furthermore, when investors expect the event to create greater return variation, the short-term implied variance rises relative to the long-term implied variance because

[3] This event-study approach distinguishes our measure from other potential measures of risk information, such as annual changes in textual attributes, changes in realized volatility or the cost of capital surrounding the announcement, or discount-rate changes derived from return decompositions (as in Vuolteenaho [2002], Penman and Zhu [2014]). By using short-horizon changes in market prices, our measure isolates only the novel information that investors learn from the event. In contrast, for example, changes in volatility surrounding an announcement could occur even when investors do not learn from the event. Similarly, discussions of risk factors may reflect sources of risk already known to investors.

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the event reflects a greater proportion of a short-term option's remaining lifetime (Dubinsky et al. [2019]). In sum, our measure, $RiskInfo_t$, equals the announcement-date change in the implied variance plus an adjustment that captures the expected decline in the implied variance.

A key appeal of our measure is that it can be easily implemented from option prices. To illustrate how it may be applied, we calculate $RiskInfo_t$ for a sample of approximately 85,000 earnings announcements from 1996 through 2019. Our findings suggest investors learn an economically significant amount of risk information from earnings announcements. Specifically, investors receive information on firms' riskiness equivalent in magnitude to approximately 29% (13%) of the implied variance in firms' stock returns expected over the month (six months) following the announcement. Moreover, our point estimates suggest the amount of risk information conveyed by earnings announcements has grown over time.

We next perform a series of validation tests on our sample of earnings announcements, demonstrate the variety of ways in which it may be used, and illustrate inference problems when using simple changes in implied volatility to assess the information in earnings. We begin by validating a central prediction of our model: $RiskInfo_t$ should positively forecast innovations in return volatility that follow earnings announcements as the risks revealed by these announcements are resolved and impounded into stock prices. Consistent with this prediction, we show $RiskInfo_t$ positively predicts within-firm changes in return volatility.

Strikingly, we also find that $RiskInfo_t$ not only forecasts changes in firms' idiosyncratic volatilities, but also forecasts changes in their sensitivities to market returns and multiple asset-pricing factors that explain the cross-section of stock returns. This suggests earnings announcements provide a diversity of information on risk. Further supporting this view, we show greater $RiskInfo_t$ forecasts decreases in firms' liquidity and increases in the discount rate investors apply to future earnings as measured by firms' implied costs of capital.

Our study also highlights inference problems when researchers use changes in implied volatility to study risk information. To illustrate the potential for Type I errors, we show implied volatilities fall at the time of earnings announcements for firms with greater analyst coverage, which may lead researchers to conclude that investors predictably overestimate risk for these firms and correct these errors during the announcements, despite coverage being observable well in advance. We also show, however, that this effect is driven by investors' anticipation of heightened announcement volatility for firms with greater analyst coverage, and thus implied volatility falls *mechanically* post announcement. Use of $RiskInfo_t$, by contrast, accounts for this mechanical change and shows no relation to analyst coverage. In addition, we highlight the potential for Type II errors using the relation between earnings surprises and risk information, underscoring the usefulness of our proposed approach.

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We also show that $RiskInfo_t$ reflects, in part, the information investors glean from earnings announcements regarding firms' financial liquidity and the scope of their risky investments. Specifically, we show higher values of $RiskInfo_t$ (which signal increased expectations of risk) foreshadow future increases in firms' spending on research and development, leverage, and distress risk.

We next test our model's prediction that $RiskInfo_t$ can be used to measure the amount of information the announcement contains on risks that will be resolved during specific time windows. For example, the difference in $RiskInfo_t$ calculated from 30- versus 60-day options captures the information the announcement contains on risks that will be resolved between 30 and 60 days following its release. Empirically, we find that these estimates successfully predict the time series of return volatility. This implies that option traders learn about the time at which these risks will be resolved and their outcomes incorporated into prices.

Additional tests show earnings announcements convey approximately five times as much risk information as nonannouncement days on average. We conduct these tests by contrasting $RiskInfo_t$ for earnings announcements versus a randomly selected "pseudo-announcement" date. These findings help justify our focus on anticipated information events and provide novel evidence that earnings announcements play a disproportionate role, relative to nonannouncement days, as conduits of risk information.

The final section of our paper applies $RiskInfo_t$ to three distinct research settings. Each setting illustrates the potential applications of $RiskInfo_t$ for future researchers while also providing platforms for discussing the potential limitations inherent in each application. These applications also collectively provide novel conceptual insights on the transmission, measurement, and implications of risk information.

In our first application, we use $RiskInfo_t$ to study the impact of the SEC's 2004 decision to require increased real-time disclosures of material changes in firms' financial condition or operations. As part of the Sarbanes-Oxley Act, the SEC required firms to disclose strategic initiatives and partnerships on a more timely basis (Noh, So, and Weber [2019]). We use a difference-in-difference design to assess how this requirement influenced the amount of risk information conveyed during firms' earnings announcements as captured by the magnitude of $RiskInfo_t$. Our findings suggest the 2004 regulation played a significant role in shaping price discovery between earnings announcements by pulling risk information forward into real-time disclosures and out of earnings announcements. We also use this application to emphasize that the appropriateness of our measure depends in part on the extent to which the information being studied is anticipated.

In our second application, we apply $RiskInfo_t$ to study the information content of common text-based proxies of risk information from firms' 10K filings (e.g., Loughran and McDonald [2011]). Our results suggest that within-firm changes in text-based proxies largely align with the *level* of firm's historical volatility but are generally

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unhelpful in forecasting changes in volatility. Text-driven proxies also fail to track variation in $RiskInfo_t$, suggesting text-based proxies more likely reflect managers acknowledging or discussing levels of risk. These tests serve to illustrate that $RiskInfo_t$ captures how investors' expectations of risk change in response to an event, but is ill-suited for studying the usefulness of textual disclosures for other purposes, for example, for studying managerial beliefs that have already been conveyed publicly through other media.

In our final application, we use S&P500 index options to study market-wide risk information conveyed by Federal Open Market Committee (FOMC) meetings. We find FOMC meetings convey substantial risk information equivalent in magnitude to approximately 21% (11%) of the implied variance in the index's return over the month (six months) following the meetings. We also find the investor sentiment index from Baker and Wurgler [2006] falls following FOMC meetings with greater $RiskInfo_t$. These findings provide novel evidence on the drivers of investor sentiment while also providing a platform for discussing the limitations of using our measure for studying market-wide events.

A central contribution of our paper is in developing an easily implementable and broadly applicable methodology. We show that researchers can measure an event's risk information as the difference between the expected and realized change in implied volatility on the event date. Our study thus builds on related studies examining option prices surrounding information events. For instance, Patell and Wolfson [1979] demonstrate option-implied volatility climbs preceding earnings announcements, and Dubinsky et al. [2019] demonstrate the magnitude of this climb predicts, with noise, realized announcement volatility. In contrast to our work, these papers assume the announcement does not contain risk information. Kelly, Pastor, and Veronesi [2016] examines index option prices surrounding anticipated political information events.

Moreover, Rogers, Skinner, and Buskirk [2009], Neururer, Papadakis, and Riedl [2016], Beatty, Cheng, and Zhang [2019], and Hann, Kim, and Zheng [2019] use changes in implied volatility surrounding earnings announcements and guidance to assess how they influence investor uncertainty. Our model illustrates that the change in investor uncertainty surrounding these announcements can be decomposed into predictable and unpredictable components. The predictable component is driven by the information in the announcement regarding expected future cash flows and the effect of earnings' information on the risk premium, whereas the unpredictable component captures the announcement's information on firm risk. Our study illustrates how isolating risk information can provide novel insights into the information provided by earnings and how simple changes in implied volatility likely entail significant measurement error as a metric of risk information.

Other work measures constructs that are related to but distinct from $RiskInfo_t$. Considerable prior work studies risk information in the financial statements using textual analysis (e.g., Loughran and McDonald

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[2011], Bao and Datta [2014], Campbell et al. [2014]). In contrast to our measure, this work focuses exclusively on qualitative risk information and does not distinguish between information in the announcement that investors learn versus already possess. Penman and Zhu [2014] and Penman and Yehuda [2019] decompose the return response to earnings into discount rate and expected cash-flow news. To quantify discount rate news, they use changes in accounting variables that predict returns (e.g., investment and accruals). This approach captures the stock-price impact of the risk information in the financial statements.[4] In contrast, our measure captures all information on stock-price risk, regardless of the degree to which it is priced; this is advantageous when testing the impact of risk information on outcomes other than equity valuation.

Our findings also build upon prior research showing option prices provide information on investors' beliefs in other contexts. For instance, Breeden and Litzenberger [1978] show option prices can be used to invert the risk-neutral return distribution, whereas Ross [2015] and Jensen, Lando, and Pedersen [2019] characterize assumptions under which option prices can be used to invert both state prices and investors' belief distribution about future returns. Borochin and Golec [2016] demonstrate option prices may be used to measure the value-effect of an event, and Smith [2018] demonstrates the change in the volatility surface leading up to a disclosure reveals its relative informativeness for good versus bad news. Kadan and Manela [2018] import index-level option prices into a structural model to derive the value of public information releases to risk-averse investors. We contribute to this line of work by helping to solve a significant measurement problem for researchers interested in studying the transmission of risk or the content of information events.

2. *Theoretical Development*

In this section, we detail how option prices can be used to measure the risk information contained in an anticipated information event. Throughout, we define the event's risk information as its impact on investors' expectations regarding the variance of future stock returns. Many types of information events may revise investors' expectations of this variance. For example, at the firm level, the balance sheet and press releases may reveal the scope of new risky investments and changes in strategy, the financial footnotes may reveal contingent liabilities such as lawsuits, and Section 7A in firms' annual reports, which discusses the magnitude of the firms' exposures to commodity and currency prices, may reveal changes in hedging practices (Kumar et al. [2008], Armstrong, Banerjee, and Corona [2013], Heinle, Smith, and Verrecchia [2018]). At the index level, news events that

[4] To interpret their metric of discount rate news as novel information to investors, one must assume changes in the financial statement return predictors are not anticipated by investors.

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relate to the levels of aggregate investment and leverage, such as changes in interest rates, may influence expected market-wide risk.

We derive our measure from a simple model of price dynamics surrounding an information event that contains information on a firm's future risk exposures (or more generally any asset's future risk exposures). The key elements of the model can be summarized as follows. Investors face uncertainty regarding the firm's forward-looking exposures to a number of risk factors. The event arrives on a known date, revealing a subset of these forward-looking exposures. In addition, on this date, the firm's price jumps by an amount that can (but need not) depend upon the change in investors' expectations of the firm's risk exposures, which captures the possibility that these exposures affect the firm's risk premium. The jump in price on the event date contains an additional stochastic component unrelated to its risk information, which reflects any information it contains on the firm's expected cash flows.

Our model takes as given the firm's stock-price process, as is the norm in the option-pricing literature, including studies of option prices surrounding information events (Dubinsky et al. [2019]). Moreover, the event in the model directly reveals information on the variance of future stock returns. This may seem at odds with the fact that information events typically do not explicitly concern the stock price but instead concern features of the firm's fundamental value. However, information on the riskiness of a firm's fundamental value naturally translates into information on the variance of its future returns. Directly modeling the price process enables a general and intuitive derivation of the measure without needing to place specific assumptions on the features of the firm's fundamentals and the precise relation between the information event and these fundamentals. In appendix A, we present an equilibrium model in which the firm's cash flows are taken as the primitive and its price endogenously follows the process similar to the one we consider in this section.

2.1 ASSUMPTIONS

Consider a firm whose stock is continuously traded over a period $[0, \infty)$. On a known date τ_D, an announcement occurs that contains information on the firm's value. Our focus is on the change in the firm's stock and option prices on the announcement date τ_D. Thus, we place no constraints on the process followed by the firm's price prior to τ_D. On date τ_D, the firm's price experiences a discrete return of $\tilde{J}$, reflecting the discrete arrival of information:

$$P_{\tau_D} = P_{\tau_D^-} \exp(\tilde{J}), \tag{1}$$

where $P_{\tau_D^-} \equiv \lim_{t \to \tau_D^-} P_t$ denotes the price just prior to the event. Following this date, the firm's stock price follows a process with stochastic volatility:

$$\frac{dP_t}{P_t} = \mu_t dt + \sum_{i=1}^{k} \tilde{\sigma}_{i,t} dW_{i,t}. \tag{2}$$

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The volatility terms $\{\tilde{\sigma}_{i,t}\}_{i\in\{1,\ldots,k\}}$ capture the firm's exposures to the k risk factors $\{W_{i,t}\}_{i\in\{1,\ldots,k\}}$. We assume that these exposures are continuous over time and that the exposures and risk factors are mutually independent.[5] Prior to date τ_D, the future values of the firm's risk exposures $\{\tilde{\sigma}_{i,t}\}_{i\in\{1,\ldots,k\}}$ are unknown to investors. The information event reveals a subset of these exposures to investors, which captures the risk information in the event. Without loss of generality, we assume that it reveals the first $j \leq k$ exposures.[6]

The jump in price $\tilde{J}$ on the event date takes the following form:

$$\tilde{J} = g(\tilde{y}, \{\tilde{\sigma}_{i,t}\}_{i\in\{1,\ldots,j\},t\in[\tau_D,\infty)}). \tag{3}$$

The first component of this jump, $\tilde{y}$, reflects information the event contains that is unrelated to its risk information and thus captures any information the event contains on expected future cash flows. We allow for a general distribution of $\tilde{y}$. The second component of the jump captures the change in the firm's discount rate that results from the risk information that the event contains.

We further allow for a generic relationship between the risk-factor exposures and the jump in price, as captured by the function $g(\cdot)$.[7] Importantly, this implies that the event can contain information on a firm's exposures to systematic risk factors, their idiosyncratic exposures, or both. The main difference between these two types of information is that, in addition to influencing investors' perceptions of future volatility, information on risk-factor exposures may also directly affect the firm's stock price on the event date by changing the risk premium. By allowing for the exposures to have different effects on the firm's price, our model captures the case in which the firm's risk premium increases in some of the firm's risk exposures (e.g., their market or value beta) but is independent of others (e.g., their exposure to idiosyncratic supply shocks).[8]

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[5] We assume that μ_t and $\{\tilde{\sigma}_{i,t}\}_{i\in\{1,\ldots,k\}}$ satisfy the usual conditions required for a well-defined solution to this equation. Moreover, it is not technically essential that the risk exposures be continuous at all points in time. We need only assume that, with probability 1, the risk exposures that are not revealed by the event do not jump on the event date. This ensures that there is no discrete change in beliefs regarding their future values on this date.

[6] Formally, let $\mathscr{F}_t^P$ and $\{\mathscr{F}_t^{\sigma_i}\}_{i\in\{1,\ldots,k\}}$ denote the filtrations generated by P_t and $\{\tilde{\sigma}_{i,t}\}_{i\in\{1,\ldots,k\}}$, respectively. Then, the market's information set at date t, which we refer to as Φ_t, equals:

$$\Phi_t = \left\{ \begin{array}{ll} \mathscr{F}_t^P & \text{for } t < \tau_D \\ \mathscr{F}_t^P \otimes \left\{\mathscr{F}_\infty^{\sigma_i}\right\}_{i\in\{1,\ldots,j\}} & \text{for } t \geq \tau_D \end{array} \right\}.$$

Although investors learn about the future values of these exposures by observing the stock price, as we will see, the precise nature of this updating is not important for the derivation of our measure.

[7] We assume that $\tilde{J}$ possesses a finite second moment in the following analysis.

[8] As a caveat, we show in appendix A that, in an equilibrium model, information on risk-factor exposures can lead to negative autocorrelation in returns, which can reduce the per-

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Our goal is to measure, for any $t > 0$, the event-date change in investors' beliefs about the future return variance over the t days that follow the event, which we refer to as $RiskInfo_t$. Formally, for any $t > \tau_D$, let $\tilde{\sigma}^2_{T,t} = \sum_{i=1}^{k} \tilde{\sigma}^2_{i,t}$ denote the total return variance. We define, for $t > 0$,

$$RiskInfo_t \equiv \underbrace{\mathbb{E}_{\tau_D}\left[\int_{\tau_D}^{\tau_D+t} \tilde{\sigma}^2_{T,s} ds\right]}_{\text{Post-event beliefs about firm risk}} - \underbrace{\mathbb{E}_{\tau_D^-}\left[\int_{\tau_D}^{\tau_D+t} \tilde{\sigma}^2_{T,s} ds\right]}_{\text{Pre-event beliefs about firm risk}}, \tag{4}$$

where the subscript τ_D^- (τ_D) denotes the information set available just prior to (after) the event. As one would expect for a notion of information arrival, under this definition, $RiskInfo_t$ has a mean of zero. Because $RiskInfo_t$ captures the change in beliefs regarding the total return variance, it encompasses both information on the firm's betas and its idiosyncratic return variance.

Given the assumption that the event reveals the first j of the k exposures, we may simplify $RiskInfo_t$ in a manner that will be useful moving forward. Let $\tilde{\sigma}^2_{D,t} = \sum_{i=1}^{j} \tilde{\sigma}^2_{i,t}$ denote the return variance driven by the exposures that the event reveals. Then, because these exposures become known after the event and no information is revealed on the remaining exposures, we have:

$$RiskInfo_t = \underbrace{\int_{\tau_D}^{\tau_D+t} \tilde{\sigma}^2_{D,s} ds}_{\text{Risk exposures revealed by the event}} - \underbrace{\mathbb{E}_{\tau_D^-}\left[\int_{\tau_D}^{\tau_D+t} \tilde{\sigma}^2_{D,s} ds\right]}_{\text{Prior expectation of these exposures}}. \tag{5}$$

We assume that there are no arbitrage opportunities such that there exists a risk-neutral measure $\mathbb{Q}$ in our model.[9] We further make the following assumption.

Assumption 1. *The risk exposures revealed by the event are firm-specific, that is, do not contain information relevant to forecasting the market-level return variance. Formally, the expectation of the return variance following the announcement is unchanged under the risk-neutral measure:* $\forall t > 0,\ \mathbb{E}^{\mathbb{Q}}_{\tau_D^-}[\int_{\tau_D}^{\tau_D+t} \tilde{\sigma}^2_{D,s} ds] = \mathbb{E}_{\tau_D^-}[\int_{\tau_D}^{\tau_D+t} \tilde{\sigma}^2_{D,s} ds]$.

Intuitively, if this assumption does not hold, investor uncertainty over the aggregate variance declines on the event date, which can lead to a decline in the variance risk premium (VRP) embedded in option contracts. Such

formance of the Black-Scholes approximation that we apply in our empirical applications. However, we conduct simulations in appendix B that show the resulting error in the measure is small under a set of parameters calibrated to our earnings-announcement sample.

[9] Note the price in our model exhibits a jump and multiple Brownian terms, which implies the market is incomplete. Thus, there may exist multiple risk-neutral measures consistent with the stock-price process. However, the expected stock-return variance is identical across these measures (Jiang and Tian [2005]), and our analysis focuses purely on this variance. Thus, we may arbitrarily select any one of these measures.

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a decline cannot readily be distinguished from a drop in investors' beliefs about the future return variance. In practical terms, the important feature is that the information in the event concerning the market-wide risk is small relative to the information it contains on the firm-level risk. This assumption appears natural when applying the measure to individual firms' announcements, as these announcements typically focus on firm-level information. However, for macroeconomic information events, the assumption will clearly be violated, and thus, our measure should be applied with caution.

Importantly, we demonstrate in the online appendix that the magnitude of the bias created by a violation of Assumption 1 may be assessed empirically in any given sample. In particular, if this restriction is violated, the event-date drop in the risk premium will create a systematic negative bias in $RiskInfo_t$, that is, $RiskInfo_t$ will have a negative mean. Thus, the mean of $RiskInfo_t$ can be used to assess the degree of bias in the sample. In section 3, we find that in the case of individual firms' quarterly earnings announcements, $RiskInfo_t$ has a mean close to zero, suggesting that the assumption is satisfied in this case.[10] We return to discuss how this bias may be minimized in the next section.

2.2 FORMAL DERIVATION

To derive the measure, we start by examining the (risk-neutral) change in the average daily return variance that is expected over the life of an option on the announcement date, which we refer to as ΔIV_t. Let $RV_{\tau_D,t}$ denote the realized return variance over t days following the disclosure date τ_D, *not including* the return variance created by the event.[11] Likewise, let $RV_{\tau_D^-,t} = \lim_{s\to\tau_D^-} RV_{s,t}$ denote the analogous return variance *including* the return variation created by information event. Then, the expected return variance over an option's remaining lifetime just prior to the event equals $\mathbb{E}^{\mathbb{Q}}_{\tau_D^-}[RV_{\tau_D^-,t}]$, and the expected return variance over an option's remaining lifetime just after the event equals $\mathbb{E}^{\mathbb{Q}}_{\tau_D}[RV_{\tau_D,t}]$. Thus, we formally define ΔIV_t as:

$$\Delta IV_t \equiv t^{-1}\left\{\mathbb{E}^{\mathbb{Q}}_{\tau_D}[RV_{\tau_D,t}] - \mathbb{E}^{\mathbb{Q}}_{\tau_D^-}\left[RV_{\tau_D^-,t}\right]\right\}, \tag{6}$$

where scaling by t places the change into variance-per-day units.

Note that ΔIV_t can be measured using the change in option prices on the announcement date. In our implementation, we approximate

[10] In untabulated tests, we further find that the mean of $RiskInfo_t$ is uncorrelated with a proxy for the extent to which a firm is likely to provide information on the market-level variance—the firm's "variance beta." In particular, we measure the firm's variance beta using the trailing one-year beta from a regression of squared daily returns on squared market returns as in Carr and Wu [2009]. This result suggests that any bias is not only small on average, but also does not meaningfully vary across firms.

[11] Formally, we let $RV_{s,t}$ denote the quadratic variation between dates s and $s+t$.

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risk-neutral expected return variances using at-the-money (ATM) Black-Scholes–implied variances; see section 3 for details. Thus, ΔIV_t simply equals the change in the Black-Scholes implied variance on the event date. Applying the Black-Scholes approach enables our measure to be applied in a simple manner to the broad cross-section of firms with liquid ATM options. However, the Black-Scholes assumptions are clearly violated in our model given the presence of stochastic volatility, which might suggest this approach would introduce noise or bias. To alleviate this concern, in appendix B, we demonstrate using simulations that, under the commonly used Cox-Ingersoll-Ross (CIR) model for the return variance, the Black-Scholes approach introduces only modest error into the measure (the average error is 6%).[12]

Conceivably, ΔIV_t is driven by the change in investors' beliefs regarding future return variance on the announcement date, and thus one might consider using it as a metric of risk information. However, ΔIV_t is a biased metric of risk information because, as originally shown by Patell and Wolfson [1979], the option-implied return variance deterministically falls following an information event. The reason is that an information event creates a discrete rise in volatility that options are subject to prior to the event but not following it. To see this formally, note:

$$RV_{\tau_D,t} = \int_{\tau_D}^{\tau_D+t} \tilde{\sigma}^2_{T,s} ds; \qquad RV_{\tau_D^-,t} = \int_{\tau_D}^{\tau_D+t} \tilde{\sigma}^2_{T,s} ds + \tilde{J}^2, \tag{7}$$

and thus, given Assumption 1, ΔIV_t reduces as follows:

$$\begin{aligned} \Delta IV_t &= t^{-1}\left\{\int_{\tau_D}^{\tau_D+t} \tilde{\sigma}^2_{D,s} ds - \mathbb{E}^{\mathbb{Q}}_{\tau_D^-}\left[\int_{\tau_D}^{\tau_D+t} \tilde{\sigma}^2_{D,s} ds\right] - \mathbb{E}^{\mathbb{Q}}_{\tau_D^-}[\tilde{J}^2]\right\} \\ &= t^{-1}\left\{RiskInfo_t - \mathbb{E}^{\mathbb{Q}}_{\tau_D^-}[\tilde{J}^2]\right\}. \end{aligned} \tag{8}$$

This equation demonstrates that ΔIV_t includes two components: (i) $RiskInfo_t$ and (ii) a deterministic drop equal to the risk-neutral expected return variance on the announcement date, $\mathbb{E}^{\mathbb{Q}}_{\tau_D^-}[\tilde{J}^2]$. Thus, to arrive at $RiskInfo_t$, we must adjust $t * \Delta IV_t$ for the variance in returns expected on the event date:

$$RiskInfo_t = t * \Delta IV_t + \mathbb{E}^{\mathbb{Q}}_{\tau_D^-}[\tilde{J}^2]. \tag{9}$$

We next show that the risk-neutral expected event-date return variance, $\mathbb{E}^{\mathbb{Q}}_{\tau_D^-}[\tilde{J}^2]$, can be estimated using the difference between the implied variance from short- and long-term options prior to the announcement. This estimate, in turn, may be added to $t * \Delta IV_t$ to arrive at $RiskInfo_t$. Intuitively,

[12] Alternatively, one could apply the model-free approach to approximating this variance (Britten-Jones and Neuberger [2000], Jiang and Tian [2005], Carr and Wu [2009]). However, in a large sample of individual-name options, many of the out-of-the-money option quotes utilized in the model-free approach are likely to be stale and noisy.

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the return variance that the event creates has a greater impact on the implied variances of short-term than long-term options, because it reflects a greater fraction of their total remaining lifetime.

To be precise, we define our estimate of $\mathbb{E}^{\mathbb{Q}}_{\tau_D^-}[\tilde{J}^2]$, $AnnVar_{t_1,t_2}$, as follows:

$$AnnVar_{t_1,t_2} \equiv \frac{t_2}{t_2 - t_1}\mathbb{E}^{\mathbb{Q}}_{\tau_D^-}\left[RV_{\tau_D^-,t_1}\right] - \frac{t_1}{t_2 - t_1}\mathbb{E}^{\mathbb{Q}}_{\tau_D^-}\left[RV_{\tau_D^-,t_2}\right]. \tag{10}$$

The terms $\mathbb{E}^{\mathbb{Q}}_{\tau_D^-}[RV_{\tau_D^-,t_1}]$ and $\mathbb{E}^{\mathbb{Q}}_{\tau_D^-}[RV_{\tau_D^-,t_2}]$ can be estimated using the preevent–implied variances from options with t_1 and t_2 days to maturity, respectively. Substituting from equation (7) and simplifying, we have:

$$\begin{aligned} AnnVar_{t_1,t_2} = {} & \frac{t_2}{t_2 - t_1}\mathbb{E}^{\mathbb{Q}}_{\tau_D^-}\left[\int_{\tau_D}^{\tau_D+t_1} \tilde{\sigma}^2_{T,s}ds\right] \\ & - \frac{t_1}{t_2 - t_1}\mathbb{E}^{\mathbb{Q}}_{\tau_D^-}\left[\int_{\tau_D}^{\tau_D+t_2} \tilde{\sigma}^2_{T,s}ds\right] + \mathbb{E}^{\mathbb{Q}}_{\tau_D^-}[\tilde{J}^2]. \end{aligned} \tag{11}$$

Under the following assumption, the first term in expression (11) roughly vanishes, so that $AnnVar_{t_1,t_2}$ approximately equals $\mathbb{E}^{\mathbb{Q}}_{\tau_D^-}[\tilde{J}^2]$.

Assumption 2. *The risk-neutral expectation of the daily variance of returns following the event is approximately constant between the event date and date $\tau_D + t_2$. Formally, $t_1^{-1}\mathbb{E}^{\mathbb{Q}}_{\tau_D^-}[\int_{\tau_D}^{\tau_D+t_1} \tilde{\sigma}^2_{T,s}ds] \approx t_2^{-1}\mathbb{E}^{\mathbb{Q}}_{\tau_D^-}[\int_{\tau_D}^{\tau_D+t_2} \tilde{\sigma}^2_{T,s}ds]$.*

An analogous condition has been used in the literature that applies the Black-Scholes model to assess the return variance created by earnings (Dubinsky et al. [2019]). As discussed in Dubinsky et al. [2019], Assumption 2 may be violated in certain cases, such as when the return variance is mean reverting and deviates from its long-run mean prior to the information event. However, across a sample of events, variability in the expected return variance following the announcement likely represents noise as opposed to a systematic source of confounding. Note that one could alternatively measure $AnnVar_{t_1,t_2}$ using the magnitude of the climb in an option's implied variance preceding the announcement, which also reflects the expected announcement-date return variance. However, our approach enables us to implement our measure using the OptionMetrics standardized option database, which renders it simpler to implement. These results lead to the following proposition, which is illustrated in figure 1.[13]

[13] It is useful to compare our results to those of Dubinsky et al. [2019], who measure the amount of return variation investors anticipate on earnings dates. They develop two measures of this variation, which they refer to as the time-series and term-structure measures. Their measures correspond directly to $-t * \Delta IV_t$ and $AnnVar_{t_1,t_2}$ in our model, respectively. Thus, $RiskInfo_t$ is precisely equal to the difference between their term-structure and time-series measures. Although, in Dubinsky et al. [2019]'s model, $-t * \Delta IV_t$ and $AnnVar_{t_1,t_2}$ are theoretically equivalent, they differ in our model precisely because investors' expectations of future variance discretely change on the announcement date. Our theoretical results thus lend support to the term-structure measure in Dubinsky et al. [2019] ($AnnVar$) as a method to capture

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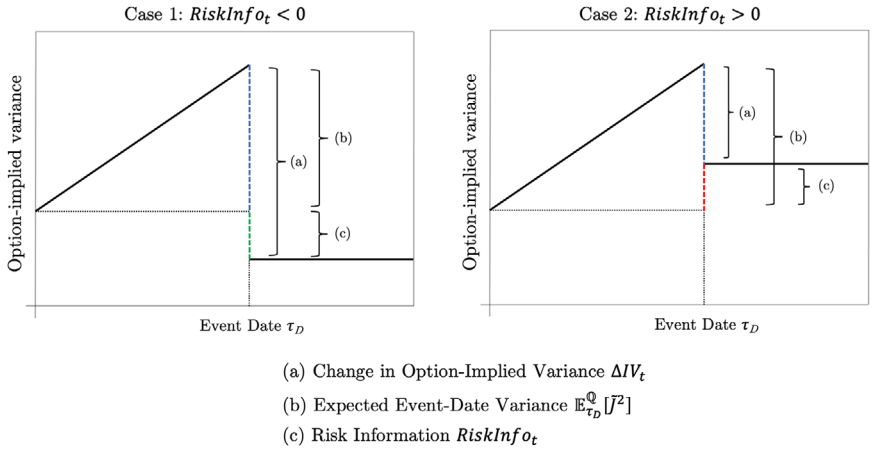

FIG 1.—This figure depicts the derivation of our measure when risk information is negative (left panel) and positive (right panel). For illustrative purposes, we assume in the plots that the event is the only source of news regarding the return variance. Thus, the implied variance climbs steadily until the announcement date, as in Patell and Wolfson [1979], and then changes on the event date by an amount that equals the difference between the event's risk information and the magnitude of this climb. As risk information is negative (positive) in the left-hand (right-hand) figure, the event-date drop in implied variance exceeds (falls short of) the initial climb in the implied variance.

Proposition 1. *Risk information of horizon t can be measured as:*

$$RiskInfo_t \approx t * \Delta IV_t + AnnVar_{t_1,t_2}, \tag{12}$$

where ΔIV_t is defined in expression (6), and $AnnVar_{t_1,t_2}$ is defined in expression (10).

Note that, under Assumption 1, the option maturities used to calculate $AnnVar_{t_1,t_2}$ may be arbitrary. However, in the online appendix, we show that one can minimize any bias created by information on the market-level variance by using an option of similar horizon to calculate both ΔIV_t and $AnnVar_{t_1,t_2}$, as we do in our main analyses (specifically, $t \approx t_1$). In this case, information on the market-level variance prior to date t does not create bias. For example, in our empirical analyses in which we calculate $RiskInfo_{30}$ using $AnnVar_{30,60}$ (i.e., 30- and 60-day options), earnings news that helps predict the market-level implied variance in the next 30 days does not produce a bias. This may contribute to our finding that our measure has a negligible mean and thus exhibits minimal bias in our earnings sample.[14]

anticipated earnings-date return variation. Their time-series measure, which corresponds to $-t * \Delta IV_t$ in our framework, also includes the risk information in earnings.

[14] This finding enables us to reconcile our result in the earnings announcement sample that $RiskInfo_{30}$ has a mean of zero and thus does not exhibit a bias with the negative short-term straddle returns observed around earnings events (Barth and So [2014]). Because such

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Note that we derive $RiskInfo_t$ under the assumption that the announcement is anticipated by investors. Uncertainty regarding the precise timing of an announcement could introduce noise into our measure but would not invalidate it. The reason is that, if investors knew an announcement was to arrive but were not certain when, $AnnVar_{t_1,t_2}$ would depend not only on the expected volatility that the announcement will create but also investors' perceived probabilities that the announcement will arrive prior to the dates t_1 and t_2. As long as these perceived probabilities do not systematically vary across events, this would lead to a reduction in the measure's power but would not create bias. For information events that are completely unanticipated by investors, ΔIV_t may be applied directly to assess the risk information in the event, without the need to adjust for $AnnVar_{t_1,t_2}$. The reason is that, for unanticipated events, investors do not impound the effect of the event on volatility into option prices prior to its release. Thus, there is no predictable drop in volatility associated with the price jump expected on the event date.

Finally, following the standard representative-agent asset-pricing approach, we do not explicitly consider disagreement among investors regarding the stock's risk. Differences in the level of disagreement across firms would not appear to influence our measure, because it is derived from changes in option prices. However, certain announcements may lead to changes in investor disagreement regarding firm risk.

In these settings, theory suggests our measure will capture how the disclosure changes the *average* investor's beliefs regarding future firm risk (where the beliefs of larger investors are weighted more heavily). For example, Smith [2019] considers a model in which investors possess different signals regarding the firm's risk, finding that the price of a derivative whose value depends upon this risk captures a weighted average of these signals. This implies, for instance, that, if an announcement simply generates or resolves disagreement over risk but does not change the average investor's beliefs about risk, our measure will roughly equal zero. Alternatively, if the announcement generates disagreement over risk because it is understood only by a subset of investors, it will capture the product of the event's risk information and the fraction of investors who understand it. When applying the measure, one should also bear in mind that it captures how the event influences the beliefs of the average investor in the option market, which may diverge from those of other stakeholders.

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straddle returns are concentrated in short-term options, they have offsetting impacts on IV_t and $AnnVar$, and thus do not impact $RiskInfo_t$. This holds regardless of whether short-term straddles earn negative returns because of a risk premium or some other channel, such as demand for leverage to speculate on earnings or demand from noise traders (Garleanu, Pedersen, and Poteshman [2008]). The precise source of price inflation in short-term straddles is not important for the efficacy of our measure because, in any case, it will be picked up and corrected for via $AnnVar$.

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2.3 THEORETICAL PREDICTIONS

We next discuss our model's predictions regarding the relation between $RiskInfo_t$ and the properties of stock returns, which we will examine in our empirical analyses. The first and most straightforward prediction from our framework is that $RiskInfo_t$ forecasts the realized variation in returns over t days following the announcement. This follows directly from the definition of $RiskInfo_t$ in expression (4) and simply captures the fact that any metric of risk information should correlate with future risk. In our empirical analysis, we focus on the measure's relationship with return volatilities, as opposed to variances, given that these volatilities are more commonly applied in the empirical literature.[15]

It is possible but not immediately obvious that the measure will forecast both a firm's idiosyncratic volatility and systematic risk-factor exposures. Although the measure captures information on both idiosyncratic risk and systematic risk exposures, certain events may contain information on only one of these types of risk. For example, the disclosure of an impending lawsuit (corporate hedging practices) would affect only perceived future idiosyncratic risk (factor exposures). Other events, such as the disclosure of a new investment, are likely to affect both sources of return variation. In our model, the measure will forecast future idiosyncratic risk and squared systematic risk exposures if and only if the event contains information on idiosyncratic and systematic sources of firm risk, respectively. Note the measure will reflect the firms' *squared* risk-factor exposures, because it is the square of these exposures that determines the firm's return variance.

If the announcement indeed contains information on a firm's systematic risk-factor exposures, conventional asset-pricing theory suggests that investors should respond to this information by adjusting the firm's discount rate. This suggests a potential relationship between $RiskInfo_t$ and the discount rate applied to a firm's expected future cash flows. However, theoretically, $RiskInfo_t$ captures information on the firm's squared risk-factor exposures, whereas signed factor exposures drive the risk premium. Nevertheless, we expect a positive association between $RiskInfo_t$ and the discount rate. The reason is that the average firm's payoffs covary positively with the market.[16] Consequently, an increase in the average firm's risk-factor exposures should have a greater effect on the firm's squared risk-factor exposures than a decrease in the average firm's risk-factor exposures.

Finally, our model predicts a relationship between the difference in $RiskInfo_t$ calculated at two different dates and the time series of future volatility. Intuitively, this difference captures the risk that investors expect will manifest in prices between these two dates. Formally, to see this, note

[15] Our results continue to hold when examining return variances as opposed to volatilities.

[16] More generally, the average firm's payoffs covary positively with the stochastic discount factor in most asset-pricing models.

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that, for $t_2 > t_1$,

$$\begin{aligned} RiskInfo_{t_2} - RiskInfo_{t_1} &= \int_{\tau_D}^{\tau_D+t_2} \tilde{\sigma}_{D,s}^2 ds - \mathbb{E}_{\tau_D^-}^{\mathbb{Q}}\left[\int_{\tau_D}^{\tau_D+t_2} \tilde{\sigma}_{D,s}^2 ds\right] \\ &\quad - \left(\int_{\tau_D}^{\tau_D+t_1} \tilde{\sigma}_{D,s}^2 ds - \mathbb{E}_{\tau_D^-}^{\mathbb{Q}}\left[\int_{\tau_D}^{\tau_D+t_1} \tilde{\sigma}_{D,s}^2 ds\right]\right) \\ &= \int_{\tau_D+t_1}^{\tau_D+t_2} \tilde{\sigma}_{D,s}^2 ds - \mathbb{E}_{\tau_D^-}^{\mathbb{Q}}\left[\int_{\tau_D+t_1}^{\tau_D+t_2} \tilde{\sigma}_{D,s}^2 ds\right]. \end{aligned} \tag{13}$$

In words, the difference in $RiskInfo_t$ calculated at the two dates relates directly to the variance in returns between those dates. We summarize these results in the following proposition.

Proposition 2.

1. *$RiskInfo_t$ positively forecasts total return volatility over the t days following the announcement.*
2. *$RiskInfo_t$ may positively forecast both idiosyncratic return volatility and the firm's squared risk-factor exposures over the t days following the announcement.*
3. *$RiskInfo_t$ is positively associated with the firm's discount rate following the announcement.*
4. *For $t_2 > t_1$, $RiskInfo_{t_2} - RiskInfo_{t_1}$ positively forecasts total return volatility from t_1 to t_2 days following the announcement.*

We empirically test these predictions in the following section.

3. Data and Results

In this section, we detail our sample, empirically implement our proposed measure of risk information, and subject it to a battery of validity tests based on our model's predictions.

3.1 SAMPLE CONSTRUCTION AND DESCRIPTIVE STATISTICS

We construct the main data set used in our analyses from four sources. We obtain price and return data from CRSP, firm fundamentals from Compustat, standardized option data from OptionMetrics, and analysts' forecasts of earnings from I/B/E/S. Our main tests center on firms' quarterly earnings announcements. Identifying the precise announcement date is important for studying short-window changes in option prices; thus, we measure earnings announcement dates using the approach taken by DellaVigna and Pollet [2009], comparing announcement dates in I/B/E/S and Compustat and using the earlier of the two if the announcement dates differ.[17]

[17] DellaVigna and Pollet [2009] show their approach yields an accuracy rate of greater than 95% and Johnson and So [2018] show these corrections help reduce bias in short-window return tests centered on earnings announcements.

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To mitigate concerns over option-market liquidity, we focus on firms with liquid options that trade an average of at least 25 contracts over the month ending five days before their earnings announcement date, though our results do not appear highly sensitive to this choice. Similarly, we eliminate firms that have been in the CRSP database for less than six months to ensure sufficient data to calculate historical return volatility and momentum. We also require that firms have coverage in the I/B/E/S database to calculate analyst-based earnings surprises and eliminate firms with prices below $1. The OptionMetrics database and hence the sample for this study spans from 1996 through 2019. Finally, we focus on firm-quarters for which we can calculate our measure of risk information using one- and six-month options, resulting in a final sample of 85,381 firm-quarter earnings announcements.

Table 1 contains descriptive statistics for key variables in our main earnings announcement sample. Panel A contains annual descriptive statistics and shows the number of observations in our sample steadily grows over time. This trend reflects the rapid growth of options trading in recent years highlighted by Johnson and So [2017], suggesting our proposed measure offers increasing coverage for the cross-section of firms over time.

Panel A also reports time-series sample averages estimated from 30-day and 182-day standardized options, denoted $RiskInfo_{30}$ and $RiskInfo_{182}$. For illustration purposes, we implement $RiskInfo_t$ for options of two horizons, though our approach can easily be modified to options of alternative horizons. For example, if researchers want to capture risk information over the next year, they can estimate $RiskInfo_t$ using 365-day options.

We implement $RiskInfo_t$ following expression (12) in section 2, employing ATM-implied variances. Consistent with the standard approach in short-window event studies, to account for the possibility of information leakage, in calculating ΔIV_t, we approximate the pre-announcement-implied variance with the implied variance at market close two days prior to the firm's earnings announcement date, $IV_{EA-2,t}$, where $EA-d$ denotes $-d$ days prior to the announcement date and the t subscript denotes the option's days to maturity.

Likewise, to account for a gradual reaction to earnings, we approximate the post-earnings–implied variance with the implied variance at the close of trading one day after the announcement, denoted $IV_{EA+1,t}$. Finally, we calculate $AnnVar$ two days prior to EA using 30- and 60-day options following the approach from Dubinsky et al. [2019]. To summarize, for example, we implement our 30-day measure of risk information, $RiskInfo_{30}$, as:

$$RiskInfo_{30} = 30 \cdot [IV_{EA+1,30} - IV_{EA-2,30}] + \frac{IV_{EA-2,30} - IV_{EA-2,60}}{\frac{1}{30} - \frac{1}{60}},$$

where the scalar for the first term, 30, reflects our use of 30-day standardized options. Finally, the denominator in the second term, $\frac{1}{30} - \frac{1}{60}$, reflects the need to estimate the option term-structure by converting daily implied variances into variances over the life of the option.

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TABLE 1
Descriptive Statistics

Panel A: Descriptive statistics

		30-Day Options			182-Day Options		
	Observations	Level $RiskInfo_{30}$	Abs. Value $RiskInfo_{30}$	Scaled $SRiskInfo_{30}$	Level $RiskInfo_{182}$	Abs. Value $RiskInfo_{182}$	Scaled $SRiskInfo_{182}$
1996	1,210	0.003	0.008	0.270	0.004	0.021	0.127
1997	1,973	0.002	0.008	0.233	0.001	0.022	0.110
1998	2,166	0.000	0.010	0.230	−0.001	0.028	0.114
1999	2,477	0.000	0.011	0.210	0.000	0.028	0.096
2000	2,260	0.008	0.019	0.227	0.011	0.045	0.094
2001	2,203	0.003	0.010	0.176	−0.001	0.025	0.073
2002	2,248	0.005	0.011	0.233	0.004	0.027	0.095
2003	2,394	−0.001	0.006	0.199	−0.003	0.016	0.093
2004	3,049	0.000	0.004	0.193	0.001	0.012	0.089
2005	3,359	0.000	0.004	0.201	0.000	0.012	0.121
2006	3,695	0.000	0.004	0.197	0.000	0.011	0.113
2007	4,211	0.001	0.005	0.304	0.006	0.016	0.128
2008	4,416	0.004	0.014	0.237	0.009	0.041	0.111
2009	4,213	−0.001	0.010	0.199	−0.003	0.032	0.102
2010	4,264	0.000	0.006	0.258	0.003	0.019	0.135
2011	4,067	0.002	0.008	0.333	0.008	0.026	0.250
2012	3,492	−0.001	0.006	0.389	0.000	0.019	0.189
2013	4,157	−0.001	0.004	0.296	0.000	0.013	0.169
2014	4,477	−0.001	0.006	0.296	0.000	0.018	0.176
2015	4,659	0.000	0.007	0.269	0.002	0.022	0.132
2016	4,628	0.000	0.009	0.315	0.000	0.029	0.129
2017	4,744	−0.001	0.006	1.122	−0.001	0.018	0.137
2018	5,435	0.000	0.009	0.425	0.002	0.023	0.135
2019	5,584	−0.001	0.010	0.362	0.001	0.023	0.208
All	85,381	0.001	0.008	0.290	0.002	0.023	0.126

Panel B: Spearman correlations

	(1)	*(2)*	*(3)*	*(4)*	*(5)*	*(6)*	*(7)*	*(8)*	*(9)*
(1) $RiskInfo_{30}$		0.931	0.669	0.643	0.490	0.050	0.022	−0.013	0.061
(2) $SRiskInfo_{30}$	0.931		0.649	0.689	0.451	0.004	0.025	0.037	0.057
(3) $RiskInfo_{182}$	0.669	0.649		0.936	0.342	0.015	−0.010	0.029	0.076
(4) $SRiskInfo_{182}$	0.643	0.689	0.936		0.339	0.017	−0.010	0.027	0.081
(5) ΔIV_{30}	0.490	0.451	0.342	0.339		0.123	0.057	−0.176	0.081
(6) $SIZE$	0.050	0.004	0.015	0.017	0.123		−0.169	−0.546	0.057
(7) LBM	0.022	0.025	−0.010	−0.010	0.057	−0.169		−0.059	−0.025
(8) $VLTY$	−0.013	0.037	0.029	0.027	−0.176	−0.546	−0.059		0.009
(9) $\Delta VLTY_{30}$	0.061	0.057	0.076	0.081	0.081	0.057	−0.025	0.009	

Panel A contains sample counts as well as time-series averages of absolute values and signed values and corresponding t-statistics for $RiskInfo_t$ using one- and six-month options around earnings announcements. $RiskInfo_t$ is our proxy for the amount of information about risk contained in a firm's disclosure. $SRiskInfo_t$ is defined as risk information scaled by diffusive variance. We calculate the average of $RiskInfo_t$ for each year in our sample. Panel B contains Spearman correlations of key variables in our analyses above (below) the main diagonal. ΔIV_{30} is change in 30-day implied variance from standardized options in the three-day window centered on firms' announcement date. $SIZE$ is the log of market capitalization, LBM is the log of firm's book-to-market ratio, $VLTY_{30}$ is the standard deviation of firm's daily returns over the 30 trading days ending five days prior to the earnings announcement, and $\Delta VLTY_{30}$ is the post-announcement change in return volatility, defined as the level difference in the standard deviation of firm's daily returns over the 30 trading days starting five days after the earnings announcement relative to $VLTY_{30}$. The sample for this analysis spans 1996 through 2019 and consists of 85,381 quarterly earnings announcements.

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Panel A of table 1 reports averages of risk information over the 30- and 182-day horizons. The first column of each side of panel A reports the averages of $RiskInfo_t$, which show both signed risk information measures have means indistinguishable from zero. Intuitively, because $RiskInfo_t$ captures "news" about firms' riskiness, we would not expect news to be on average positive or negative. This finding provides evidence that the assumption used to derive the measure that the event contains minimal information on market-level risk (Assumption 1) is satisfied in our sample.

Panel A also reports that the average absolute values of $RiskInfo_t$ are 0.008 over the 30-day horizon and 0.023 over the 18-day horizon. These figures indicate the average impact of earnings announcements on investors' expectations of the future return variance. To illustrate the economic significance of these effects, we also report measures of risk information, denoted $SRiskInfo_{\tau_M}$ that scale the average absolute values of $RiskInfo_{30}$ and $RiskInfo_{182}$ by firms' diffusive return variance, where diffusive variance reflects the units of variance expected over the remaining life of the option.[18]

Using this scaled approach, our summary statistics in panel A indicate investors update on firms' riskiness by approximately 29% of the expected variance of stock returns over the 30 days following the announcement and 13% of the variance expected over the 182 days following the announcement. These sample averages suggest investors receive an economically large amount of information on firms' risks from earnings announcements. Moreover, the annual time series suggests the amount of risk information that firms convey via their earnings announcements gradually increased over our sample period.

Panel B of table 1 presents Spearman correlations of our main variables. $RiskInfo_{30}$ has a positive correlation of 0.49 with the change in implied variance from 30-day options, ΔIV_{30}. $RiskInfo_{30}$ also relates negatively to firms' return volatility over the 30 days ending five days prior to the announcement, $VLTY_{30}$, and positively to $\Delta VLTY_{30}$, which reflects the difference in return volatility over the 30 days post- versus pre-announcement. Panel C also shows the two measures of $RiskInfo_t$ are highly correlated across 30- and 182-day options, suggesting the risk information we measure is persistent and our 30-day version likely captures information on fairly persistent components of firms' riskiness.

3.2 VALIDATION TESTS: FORECASTING VOLATILITY

Our first main tests examine the ability of $RiskInfo_t$ to predict changes in return volatility. Specifically, in panel A of table 2, we sort firms into quintile portfolios based upon the distributional breakpoints of $RiskInfo_{30}$ from the prior calendar quarter. Then, we calculate the average changes

[18] We calculate diffusive variance following the approach of Dubinsky et al. [2019], which uses the term-structure of implied volatility prior to earnings announcements to remove the influence of heightened volatility associated with firms' earnings announcements.

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TABLE 2
Validation Tests

Panel A: Averages changes in total volatility

	Q1 (Low)	Q2	Q3	Q4	Q5 (High)	High-Low
$\Delta VLTY_{30}$	−0.314	−0.169	−0.140	−0.147	−0.156	0.158
	−(7.68)	−(5.49)	−(4.54)	−(4.48)	−(3.05)	(5.62)

Panel B: Regressions of changes in volatility

	(1)	(2)	(3)	(4)	(5)	(6)
$RiskInfo_{30}$	5.930***	5.832***	5.799***	–	–	4.202***
	(8.87)	(8.88)	(8.87)	–	–	(6.81)
ΔIV_{30}	–	–	–	3.394***	3.649***	2.256***
	–	–	–	(5.39)	(5.32)	(3.70)
SIZE	–	0.052***	0.053***	0.055***	0.054***	0.052***
	–	(3.17)	(3.16)	(3.36)	(3.31)	(3.14)
LBM	–	−0.254***	−0.259***	−0.259***	−0.259***	−0.258***
	–	(−3.95)	(−3.94)	(−4.03)	(−4.01)	(−4.02)
SURP	–	–	−1.470	−1.643	−1.604	−1.487
	–	–	(−0.99)	(−1.12)	(−1.09)	(−0.99)
AnnVar	–	–	–	–	1.438	
	–	–	–	–	(1.23)	
Obs	84,735	84,735	84,735	84,735	84,735	84,735
Adj. R^2	0.073	0.075	0.075	0.075	0.075	0.077
Firm FE?	Y	Y	Y	Y	Y	Y
Year FE?	Y	Y	Y	Y	Y	Y

Panel C: Subsample analysis

Subsample:	Low DVOL (1)	Mid DVOL (2)	High DVOL (3)	Non-S&P500 (4)	S&P500 (5)
$RiskInfo_{30}$	2.999***	6.383***	8.198***	5.545***	8.285***
	(3.48)	(5.90)	(5.04)	(7.88)	(4.13)
Obs	27,191	30,729	25,285	58,216	26,484
Adj. R^2	0.087	0.109	0.111	0.068	0.068
Controls?	Y	Y	Y	Y	Y
Firm FE?	Y	Y	Y	Y	Y
Year FE?	Y	Y	Y	Y	Y

Panel D: Decomposition of volatility

Dep. Variable:	$\Delta IVOL_{30}$ (1)	$\Delta(\beta^2_{MKT})$ (2)	$\Delta(\beta^2_{SMB})$ (3)	$\Delta(\beta^2_{HML})$ (4)	$\Delta(\beta^2_{UMD})$ (5)
$RiskInfo_{30}$	0.343***	8.854**	12.006**	23.126***	3.671
	(7.78)	(2.62)	(2.80)	(2.81)	(0.91)
Obs	84,748	84,748	84,748	84,748	84,748
Adj. R^2	0.081	0.072	0.059	0.058	0.076
Controls?	Y	Y	Y	Y	Y
Firm FE?	Y	Y	Y	Y	Y
Year FE?	Y	Y	Y	Y	Y

Panel A contains average changes in, and corresponding t-statistics for, firms' volatility relative to their earnings announcement date, sorted into quintile portfolios of $RiskInfo_{30}$. $RiskInfo_{30}$ is our proxy for the

(*Continued*)

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TABLE 2—*(Continued)*

amount of information about risk contained in a firm's disclosure from 30-day options. Quintile portfolios are formed on a rolling basis, using the distributional breakpoints of $RiskInfo_t$ from the prior calendar quarter. High-Low denotes the difference between the first and fifth quintile. $\Delta VLTY_{30}$ is the post-announcement change in return volatility, defined as the difference in the standard deviation of firm's daily returns over the 30 days starting five days after the earnings announcement, relative to the standard deviation of firm's daily returns over the 30 days ending five days prior to the earnings announcement, and multiplied by 100. *SIZE* is the log of market capitalization, *LBM* is the log of firm's book-to-market ratio, and *SURP* is the firms' analyst-based surprise at their earnings announcement. ΔIV_{30} is the change in the 30-day implied variance from standardized options in the three-day window centered on firms' announcement date. $AnnVar$ equals the anticipated level of volatility at the earnings announcement derived from option prices following Dubinsky et al. [2019]. Firm and year fixed effects are used throughout. The first three columns of panel C contain analogous tests for subsamples of firms based on their option trading volume. "Low DVol," "Mid DVol," and "High DVol" indicate whether the option is in the lowest, middle, or highest subsample of option dollar trading volume over the month ending five days before firms' earnings announcements, where subsamples are formed each calendar quarter. The remaining two columns contain analogous tests for subsamples of firms in the S&P500 versus non-S&P500 firms. Panel D reports regressions when decomposing return volatility in changes in firms' idiosyncratic volatility versus changes in firm-level factor loadings. We again calculate changes as the difference over the 30 days starting 5 days after firms' earnings announcement relative to the 30 days ending five days prior to firms' earnings announcements. We measure changes in idiosyncratic volatility ΔIVOL_{30} by taking the squared residual from a daily four-factor model whereas we measure the changes in factor loadings from the same regression as follows: $\Delta(\beta^2_{MKT})$ is the change in firms' squared sensitivity to the market factor; $\Delta(\beta^2_{SMB})$ is the change in firms' squared sensitivity to the small-minus-big factor; $\Delta(\beta^2_{HML})$ is the change in firms' squared sensitivity to the high-minus-low book to market factor; and $\Delta(\beta^2_{UMD})$ is the change in firms' squared sensitivity to the up-minus-down momentum factor available from Ken French's Web site. The parentheses contain *t*-statistics based on standard errors clustered by firm and year. The sample for this analysis spans 1996 through 2019 and consists of 85,381 quarterly earnings announcements.

in firms' volatilities relative to their earnings announcement dates for each portfolio as the change in post-to-pre volatility.

In panel A, to match our use of 30-day options when calculating $RiskInfo_{30}$, we measure changes in volatility as the standard deviation of firm's daily returns over the 30 days starting five days after the earnings announcement minus the standard deviation of firm's daily returns over the 30 days ending five days prior to the earnings announcement.[19] In measuring volatility, we conservatively skip the days immediately following the announcement to avoid issues with the gradual incorporation of information into prices. The column labeled "High-Low" denotes the difference between the first and fifth quintile.

Panel A of table 2 shows that changes in volatility monotonically increase across quintiles of $RiskInfo_{30}$. The spread in $\Delta VLTY_{30}$ across the highest and lowest quintile is 0.16 (t-statistic = 5.62), which is large relative to the sample average of the absolute pre-to-post change in return volatility surrounding earnings announcements of 0.18.[20] These findings help validate our measure and are consistent with our model's prediction that investors glean

[19] Our results are qualitatively identical if we study log changes in volatility rather than level changes or skip fewer days surrounding the announcement when measuring volatility (results untabulated).

[20] The observation that volatility drops on average following the announcements is consistent with Brandt et al. [2009], who find that return volatility has declined in recent decades.

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information about firms' riskiness from their earnings announcements and incorporate this information into option prices.

For most of our analyses, we focus on $RiskInfo_{30}$ for consistency and because short-term options tend to be more actively traded. However, we find our results are not highly sensitive to using options of longer horizon options, which is consistent with the evidence in panel C of table 1 that the risk information we measure concerns persistent sources of risk.

Panel B of table 2 examines the incremental predictive power of $RiskInfo_{30}$ in multivariate tests. These tests include firm and year fixed effects to explore within-firm variation and mitigate the influence of secular trends across time-periods. The parentheses contain t-statistics based on standard errors clustered by firm and year. Columns 1, 2, 3, and 6 show the predictive power of $RiskInfo_{30}$ for changes in volatility is highly robust and incremental to standard controls known to explain variation in return volatility: firm size, the log book-to-market ratio, and the analyst-based earnings surprise (e.g., Sridharan [2015]). These results further suggest that the risk information in earnings announcements is not fully captured by the earnings surprise itself. Thus, the financial statements, performance metrics, and qualitative discussion that accompany with these announcements appear to contain significant risk information.

Columns 4 and 5 confirm that simple changes in implied variance, ΔIV_{30}, around the announcement help forecast changes in volatility when not controlling for $RiskInfo_{30}$. Column 5 also shows $AnnVar$ positively predicts changes in volatility when controlling for ΔIV_{30} in specifications with firm-fixed effects, which is consistent with its ability to correct for measurement error in ΔIV_{30} as a metric of risk. Column 6 indicates that $RiskInfo_t$ offers incremental predictive power for volatility over ΔIV_{30} by isolating how investors revise their priors over future return variance in response to an information event.[21]

Panel C of table 2 shows our results are robust across several subsamples partitioned by option liquidity. The notation in the bottom row of columns 1 through 3, "Low DVOL," "Mid DVOL," and "High DVOL," indicates whether the option is in the lowest, middle, or highest tercile of option dollar trading volume over the month ending five days before firms' earnings announcements, where subsamples are formed each calendar quarter. Columns 4 and 5 contain analogous tests of subsamples of S&P500 and non-S&P500 firms. These tests show our inferences are robust across all subsamples and unlikely to be driven by illiquid options or limited to firms

[21] Column 6 also shows ΔIV_{30} forecasts incremental to $RiskInfo_t$. To understand this, note that, controlling for $RiskInfo_t$, increases in ΔIV_{30} reflect decreases in $AnnVar$. In untabulated tests, we find that changes in volatility are weakly decreasing across quintile portfolios sorted by $AnnVar$. This suggests that, when investors expect a larger absolute movement in price upon earnings' release, realized volatility following earnings tends to decline relative to volatility prior to earnings. This may arise because, when more information is revealed by earnings, investors subsequently face less uncertainty over the firm's value.

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with the most liquid options. They also suggest that our measure may be applied in the broader cross-section of firms.

The remainder of table 2 examines our model's prediction that the predictive power of $RiskInfo_{30}$ for future volatility could stem either from idiosyncratic risk information, systematic risk information, or both. Column 1 of panel D of table 2 reports regression results of changes in firms' idiosyncratic volatility over the 30 days starting five days after the earnings announcement, relative to the 30 days ending five days prior to the earnings announcement. We measure idiosyncratic volatility for each date by taking the squared residual from a firm-specific market-model regression with three lags. The positive relation between $RiskInfo_{30}$ and changes in idiosyncratic risk suggests part of the predictive power of $RiskInfo_{30}$ for total returns stems, in part, from investors learning about firm-specific risks.

The remaining columns of table 2 provide evidence that $RiskInfo_{30}$ also reflects information on firms' risk-factor loadings, for example, by reflecting changes in the sensitivity of a firm's business to macroeconomic conditions. Specifically, we run regressions where the dependent variable measures pre- versus-post announcement changes in squared risk-factor loadings from a four-factor model estimated in the 30 days immediately before versus after firms' earnings announcement. We examine the change in the square of these factor loadings, as squared loadings determine the variance of a firm's returns. $\Delta(\beta^2_{MKT})$, $\Delta(\beta^2_{SMB})$, $\Delta(\beta^2_{HML})$, and $\Delta(\beta^2_{UMD})$ denote the change in firms' squared sensitivity to the market factor, small-minus-big factor, high-minus-low book-to-market factor, and up-minus-down momentum factor, respectively. Columns 2 through 5 of panel C of table 2 show that $RiskInfo_t$ forecasts changes in firms' sensitivities to the market portfolio and multiple asset-pricing factors known to explain the cross-section of stock returns. By highlighting the relation between $RiskInfo_{30}$ and both idiosyncratic risks and risk-factor exposures, the collective results in panel D illustrate that earnings announcements provide a diversity of information on firms' risk profiles.

3.3 ILLUSTRATION OF METHODOLOGICAL IMPLICATIONS

In this section, we devise tests to highlight potential inference problems when researchers use changes in implied volatility to study risk information conveyed by earnings announcements. Our model shows that this inference problem arises because investors rationally anticipate that earnings announcements elicit a jump in the stock price and thus create a short-lived increase in the return variance. This anticipation causes implied variances to predictably rise in advance of the event and mechanically decline on average after the event (Patell and Wolfson [1979], Dubinsky et al. [2019]). Because of this mechanical on-average decline, the pre-to-post announcement change in implied variance around the event exhibits measurement error when applied as a metric of risk information.

In table 3, we devise simple tests to illustrate the potential for both Type I and II errors, though an important caveat should be kept in mind.

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TABLE 3
Methodological Implications

	Dep. Variable:	ΔIV_{30}	*AnnVar*	$RiskInfo_{30}$	ΔIV_{30}	*AnnVar*	$RiskInfo_{30}$
	Intended Example	(1)	(2)	(3)	(4)	(5)	(6)
COV	*Type I Error*	−0.001*	0.001***	−0.000	−0.001*	0.001***	0.000
		(−1.72)	(3.88)	(−0.01)	(−1.69)	(3.90)	(0.08)
SURP	*Type II Error*	–	–	–	−0.022	−0.004	−0.041***
		–	–	–	(−1.13)	(−0.72)	(−3.27)
Obs		81,642	81,642	81,642	81,642	81,642	81,642
Adj. R^2		0.083	0.152	0.088	0.083	0.152	0.089
Firm FE?		Y	Y	Y	Y	Y	Y
Year x NYSD FE?		Y	Y	Y	Y	Y	Y

This table contains results from a regression of $RiskInfo_{30}$ and its components on firms' analyst-based earnings surprises ($SURP$) and the log of one plus the number of analysts covering a firm (COV). ΔIV_{30} equals the change in 30-day implied variance from standardized options in the three-day window centered on firms' announcement date. $AnnVar$ equals the anticipated level of volatility at the earnings announcement derived from option prices following Dubinsky et al. [2019]. $RiskInfo_{30}$ is our proxy for the amount of information about risk contained in a firm's disclosure from 30-day options, which equals the sum of ΔIV_{30} and $AnnVar$. In columns 1 through 3, we separately regress ΔIV_{30}, $AnnVar$, and $RiskInfo_{30}$ on COV along with firm and year-by-NYSE-size-decile fixed effects indicated at the bottom of the table. In columns 4 through 6, we run similar regressions but add $SURP$ as an independent variable. The inclusion of COV in these tests is intended to illustrate a likely Type I error in which the researcher incorrectly rejects the null when using ΔIV_{30} instead of $RiskInfo_{30}$. The inclusion of $SURP$ in these tests is intended to illustrate a likely Type II error in which the researcher incorrectly fails to reject the null when using ΔIV_{30} instead of $RiskInfo_{30}$. The parentheses contain t-statistics based on standard errors clustered by firm and year. The sample for this analysis spans 1996 through 2019 and consists of 85,381 quarterly earnings announcements.

Specifically, because we do not observe the true data-generating process, our tests instead rely on theory to infer likely inference errors. For example, in our first illustration, we rely on theory that investors rationally anticipate predictable patterns in volatility around earnings announcements. Because investors are rational, they do not systematically over- or underestimate volatility patterns. In table 3, we separately run regressions of ΔIV_{30}, $AnnVar$, and $RiskInfo_{30}$ on selected covariates, along with firm and year-by-size-decile fixed effects.

To illustrate the potential for Type I errors, we show in table 3 that implied variances fall at the time of the announcement for firms with greater analyst coverage, as indicated by the negative coefficient in column 1 on our analyst coverage measure COV. This negative coefficient may lead researchers who rely on ΔIV_{30} to study risk information to conclude that investors predictably overestimate risk for firms with high coverage and correct these expectation errors at the time of the announcement. However, this conclusion would likely be puzzling because analyst coverage is observable well in advance of the announcements, and thus we would not expect investors to systematically overestimate the amount of risk information during the earnings announcements of high coverage firms.

Column 2 of table 3 shows a strong positive relation between COV and $AnnVar$, indicating investors anticipate heightened announcement

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volatility for firms with greater analyst coverage.[22] This relation is important, as it indicates that the negative relation between COV and ΔIV_{30} is likely driven by the implied variance falling mechanically post-announcement for high coverage firms as the event-driven volatility subsides. Use of $RiskInfo_{30}$, by contrast, accounts for this mechanical change. Consistent with our approach reducing Type I errors, column 3 shows no relation between $RiskInfo_{30}$ and analyst coverage.

In columns 4 through 6 of table 3, we seek to highlight the potential for Type II errors when using ΔIV_{30} by studying the relation between earnings surprises and risk information. These tests are guided by the idea that positive earnings news should cause investors to lower their beliefs about a firm's risk profile (e.g., by lowering their beliefs over the potential for default). Consistent with this view, we find a significant negative relation between firms' analyst-based earnings surprises, $SURP$, and $RiskInfo_{30}$. By contrast, we find no significant relation between $SURP$ and ΔIV_{30}, which is likely driven by the measurement error in ΔIV_{30} as a metric of risk information.

3.4 MARKET IMPLICATIONS AND FIRMS' FUNDAMENTALS

Together with Proposition 2, the finding in section 3.2 that $RiskInfo_{30}$ captures information on firms' future risk-factor exposures suggests the discount rate investors apply to firms' cash flows should rise with $RiskInfo_{30}$. In figure 2 and table 4, we test this prediction by studying firms' implied cost of capital (ICCs) using the estimation approach of Gebhardt, Lee, and Swaminathan [2001], which reconciles analysts' earnings forecasts with firms' prevailing stock prices.[23]

An appealing feature of ICCs is that they relate prevailing market prices with the analysts' most recent expectations of earnings and thus are less likely to reflect cross-sectional variation in news about the level of firms' future cash flows (Lee, So, and Wang [2020]). To estimate how investors update discount rates in response to earnings news, we calculate changes in ICCs from the months before versus after firms' earnings-announcement dates.

Figure 2 plots the estimated relation (and 95% confidence interval) between $RiskInfo_{30}$ and changes in ICCs after accounting for firm and

[22] The heightened announcement volatility for firms with greater coverage is consistent with prior evidence that greater competition among analysts spurs timely forecast and recommendation revisions, often occurring in conjunction with earnings announcements (e.g., Cooper, Day, and Lewis [2001], Altınkılıç and Hansen [2009], Altınkılıç, Balashov, and Hansen [2013]). Thus, analyst coverage relates positively to absolute earnings announcement returns because of a greater concentration of information production around the announcements. We verify that the relationship between the earnings-date return variance and analyst coverage is positive in our sample in the online appendix.

[23] We eliminate observations with negative implied cost of capital to mitigate measurement error and facilitate the interpretation of the results.

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TABLE 4

Forecasting Changes in Market Outcomes and Fundamentals

Panel A: Forecasting market outcomes

	ΔICC		$\Delta AMIHUD$		$\Delta VW\text{-}Spreads$	
	(1)	(2)	(3)	(4)	(5)	(6)
$RiskInfo_{30}$	10.481***	9.000***	2.599***	2.109***	1.352***	1.006***
	(7.86)	(5.08)	(7.03)	(5.01)	(4.74)	(2.91)
ΔIV_{30}	–	1.412	–	−0.094	–	0.158
	–	(1.33)	–	(−0.46)	–	(1.20)
Obs	64,600	64,600	84,747	84,735	83,620	83,608
Adj. R^2	0.041	0.085	0.058	0.100	0.027	0.038
Controls?	N	Y	N	Y	N	Y
Firm FE?	Y	Y	Y	Y	Y	Y
Year FE?	Y	Y	Y	Y	Y	Y

Panel B: Forecasting changes in fundamentals

	$\Delta R\&D\ Spending$		$\Delta Leverage$		$\Delta Altman\ Z\text{-}Score$	
	(1)	(2)	(3)	(4)	(5)	(6)
$RiskInfo_{30}$	0.006***	0.007***	0.039***	0.037***	−6.156***	−5.918***
	(3.97)	(3.70)	(3.39)	(2.97)	(−4.25)	(−4.39)
ΔIV_{30}	–	-0.001	–	0.003	–	0.359
	–	(−0.63)	–	(0.37)	–	(0.69)
Obs	72,386	72,386	72,357	72,357	63,158	63,158
Adj. R^2	0.008	0.009	0.055	0.062	0.043	0.065
Controls?	N	Y	N	Y	N	Y
Firm FE?	Y	Y	Y	Y	Y	Y
Year FE?	Y	Y	Y	Y	Y	Y

Panel A contains regression results of changes in market outcomes. The first two-columns measure the post- versus pre-announcement change in firms' implied cost of capital (ICC) using the estimation approach in Gebhardt, Lee, and Swaminathan [2001] that reconciles analysts' forecasts with the current stock price. We calculate the change in ICCs from the month after firms' earnings announcement date relative to the month prior to firms' earnings announcement date. Panel A also contains regression results based on within-firm changes in firms' Amihud illiquidity ratio ($\Delta AMIHUD$), defined as absolute returns scaled by dollar trading volume, and volume-weighted bid-ask spreads ($\Delta VW\text{-}Spreads$). We measure both $\Delta AMIHUD$ and $\Delta VW\text{-}Spreads$ from the month after firms' earnings announcement date relative to the month prior to firms' earnings announcement date. Panel B contains regression results of changes in firms' subsequently reported fundamentals: changes in research and development spending scaled by total assets ($RDSpending$), debt-to-equity ratio ($Leverage$), and firms' distress risk as measured by firm's Altman Z score ($AltmanZ - Score$), where higher values indicate a lower likelihood of financing constraints. $RiskInfo_{30}$ is our proxy for the amount of information about risk contained in a firm's disclosure from 30-day options. In some specifications, we control for the following variables: *SIZE* is the log of market capitalization, *LBM* is the log of firm's book-to-market ratio, and *SURP* is the firms' analyst-based surprise at their earnings announcement. ΔIV_{30} is the change in 30-day implied variance from standardized options in the three-day window centered on firms' announcement date. Additionally, in columns 4 and 6 of panel A we add an additional control for the post-announcement change in return volatility ($\Delta VLTY_{30}$), defined as the difference in the standard deviation of firm's daily returns over the 30 days starting five days after the earnings announcement, relative to the standard deviation of firm's daily returns over the 30 days ending five days prior to the earnings announcement. Firm and year fixed effects are used throughout. The parentheses contain *t*-statistics based on standard errors clustered by firm and year. The sample for this analysis spans 1996 through 2019 and consists of 85,381 quarterly earnings announcements.

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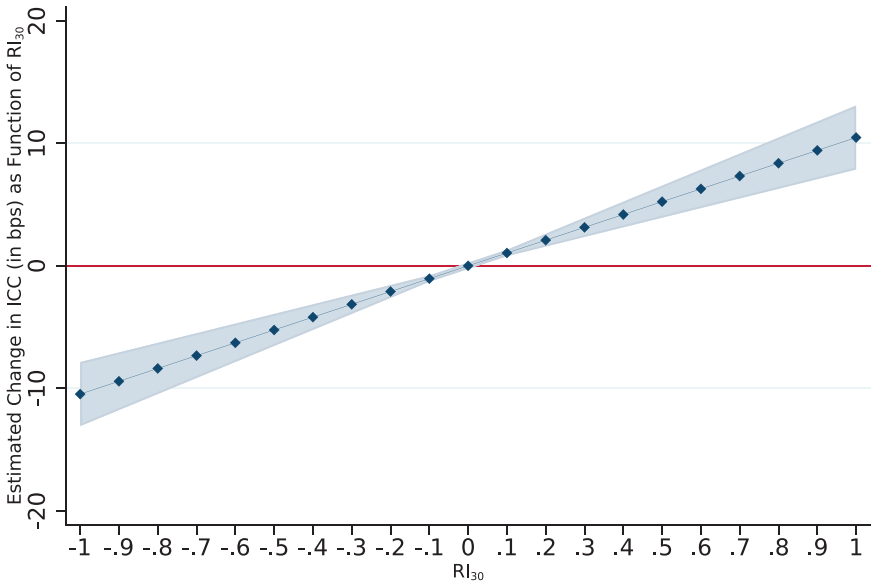

Fig 2.—Changes in implied costs of capital. This figure plots conditional estimates of within-firm changes in the post- versus pre-earnings announcement change in firms' implied cost of capital (ICC) as a function of $RiskInfo_{30}$. The graph depicts the fitted values and 95% confidence intervals stemming from pooled univariate regressions with firm and period fixed effects. We estimate ICCs using the estimation approach in Gebhardt, Lee, and Swaminathan [2001] that reconciles analysts' forecasts with the current stock price. We calculate the change in ICCs (shown in basis points) as the level difference from the month after firms' earnings announcement date relative to the month prior to firms' earnings announcement date. $RiskInfo_{30}$ is our proxy for the amount of information about risk contained in a firm's disclosure from 30-day options. Quintile portfolios are formed on a rolling basis, using the distributional breakpoints of $RiskInfo_{30}$ from the prior calendar quarter.

year fixed effects. We find greater values of $RiskInfo_{30}$ help explain post-announcement increases in ICCs. Specifically, the positive relation highlighted in figure 2 is consistent with investors discounting firms' future cash flows when their expectations of cash-flow risk increase. In terms of economic magnitudes, figure 2 indicates that pre- versus post-announcement changes in ICCs are roughly 22 basis points higher for the highest versus lowest quintile of $RiskInfo_{30}$. These findings indicate that firm disclosures not only convey information about firms' idiosyncratic risks, but also convey information about (conditional) covariances and thus, consistent with theory, are relevant to a cost of capital (Lambert, Leuz, and Verrecchia [2007], Heinle, Smith, and Verrecchia [2018]).

Regression results in columns 1 and 2 in panel A of table 4 corroborates these results in multivariate tests with standard controls. We find that $RiskInfo_{30}$ subsumes the predictive power of simple changes in implied

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variance around the announcement, consistent with ΔIV_{30} being a noisy measure of risk information gleaned from earnings announcements.[24]

Columns 3 through 6 of panel A show $RiskInfo_{30}$ also helps forecast within-firm changes in liquidity as measured by firms' Amihud illiquidity ratios ($\Delta AMIHUD$) and volume-weighted bid-ask spreads ($\Delta VW\text{-}Spreads$).[25] Both proxies are measured as the pre- to postchange in the months around the announcement date. These findings are consistent with $RiskInfo_{30}$ predicting changes in liquidity by capturing changes in the inventory risks faced by liquidity providers. Moreover, this finding is consistent with models of information acquisition, in which elevated levels of uncertainty create opportunities for sophisticated investors to acquire and trade on private information (e.g., Mendelson and Tunca [2004], Smith [2020]).

Our results thus far establish that $RiskInfo_{30}$ forecasts changes in the capital market that are consistent with traders learning about the resolution of risk from firms' earnings announcements. We next extend these findings by linking our measure to changes in firms' subsequently reported behavior and fundamental performance.

Given that earnings announcements are often accompanied by financial statements, performance metrics, and qualitative discussion, investors can likely use these announcements to learn about risks that will manifest in firms' future fundamentals. Thus, we expect that $RiskInfo_{30}$ helps forecast changes in firms' subsequent fundamental performance. Panel B of table 4 contains three sets of regression results where the dependent variables reflect forward changes in firms' investments and leverage in the fiscal quarter after our measurement of $RiskInfo_{30}$.

In panel B, the dependent variable for columns 1 and 2 is the next quarter's change in firms' research and development expenditure scaled by total assets, *FutureRND*, where we measure the change from the current period to the next fiscal quarter after our measurement of $RiskInfo_{30}$. Our findings suggest $RiskInfo_{30}$ is greater in anticipation of increased risky investments as proxied by spending on research and development.

Panel B of table 4 also contains regression results on future changes in firms' leverage ratio as measured by total liabilities to shareholders' equity and the Altman distress score. We include the Altman score as a summary metric, because we expect investors' expectations of risk to rise in response to declines in working capital as a percentage of assets, lower profitability

[24] In our online appendix, we also find that RI relates negatively to contemporaneous returns measured surrounding the earnings announcement date. This evidence is consistent with investors perceptions of risk increasing when firms report bad news.

[25] In columns 4 and 6 of panel A table 4, we add an additional control for the post-announcement change in return volatility ($\Delta VLTY_{30}$) to account for changes in liquidity that are not mechanically driven by increases in return volatility. We continue to find a significant positive relation between our measure and within-firm changes in illiquidity, consistent with our results reflecting an information effect rather than a pure volatility effect.

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or sales, or higher liabilities relative to the market value of equity, all of which are direct inputs to the calculation of the Altman score. In panel B, we find greater risk information during firms' announcements portends increases in their leverage and distress risk, consistent with investors learning about impending changes in firms' capital structure and solvency from their earnings announcements.

3.5 HORIZON OF VOLATILITY

In table 5, we examine an additional implication of our model detailed in Proposition 2: the difference between $RiskInfo_{t_1}$ and $RiskInfo_{t_2}$ should capture the amount of risk investors expect to realize in prices between t_1 and t_2 days following the announcement. Consistent with this prediction, panel A of table 5 shows differences in $RiskInfo_t$ across shorter and longer dated options predict the timing of heightened return volatility. We implement these tests by first by assigning firms to portfolios based on $\Delta RiskInfo_t$, which reflects the difference in risk information derived from 30-day and 182-day options and is defined as $RiskInfo_{30}$ minus $RiskInfo_{182}$. We then test whether $\Delta RiskInfo_t$ predicts within-firm variation in volatility in 30- versus 182-day windows following the event.

Our tests in table 5 focus on the time series of post-announcement volatility to capture how return volatility varies with time. Specifically, we measure *VolTime* as the standard deviation of firm's daily market-adjusted returns from five to 35 days after the earnings announcement minus the standard deviation of firm's daily returns from 36 to 187 days after the earnings announcement. We skip five days afterward to align with our tests in table 2. We also calculate a standardized version, $VolTime_{STND}$, to account for potential differences in the measure's power to forecast volatility across the two windows of measurement. We do so by standardizing the two return volatility measures each calendar quarter to have a mean zero and unit standard deviation and then calculate $VolTime_{STND}$ as the level difference in the standardized versions.

Panel A of table 5 shows $\Delta RiskInfo_t$ positively predicts variation in both measures of the time series of volatility. When $RiskInfo_{30}$ is high relative to $RiskInfo_{182}$, post-announcement volatility tends to be higher in the 30 days immediately following the announcement, compared to the six months following the announcement. Panel B shows this relation is robust to standard controls as well as the difference in implied variance for 30- and 182-day options denoted *IVTime*. These findings suggest option traders not only learn about total risk information from firms' earnings announcements, but also glean information about the horizon at which these risks will be resolved and their outcome incorporated into prices.

Figure 3 helps summarize a consistent theme from our tests so far that $RiskInfo_{30}$ often subsumes the predictive power of ΔIV_{30} when both

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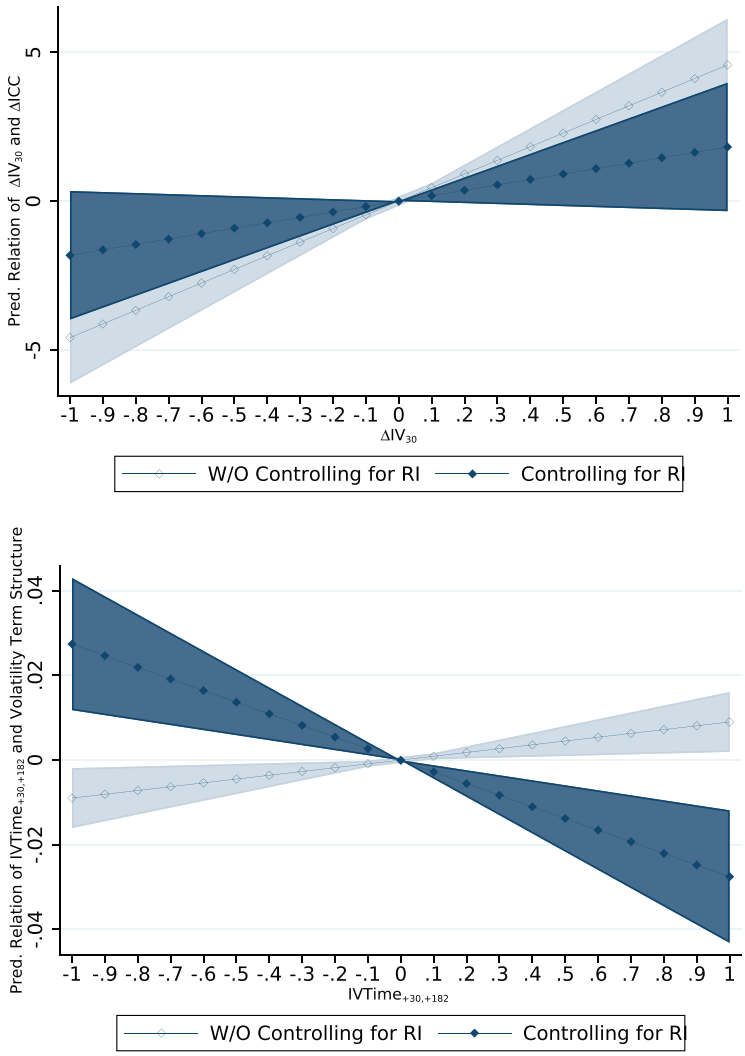

FIG 3.—Conditional estimated relations with option implied variance. This chart contains the conditional estimated predicted relations between implied volatilities with (i) changes in ICCs, and (ii) the term-structure of post-announcement volatility. The light blue portions depict the fitted values and 95% confidence intervals stemming from pooled univariate regressions with firm and period fixed effects. The dark blue portions depict the fitted values and 95% confidence intervals stemming from pooled multivariate regressions that add controls for *RiskInfo* with firm and period fixed effects. ΔICC is the change in ICCs from the month after firms' earnings announcement date relative to the month prior to firms' earnings announcement date. ΔIV_{30} is the change in 30-day implied variance from standardized options in the three-day window centered on firms' announcement date. *IVTime* is the level difference of the change in 30-day implied variance relative to the change in 182-day implied variance surrounding the earnings announcement. The sample for this analysis spans 1996 through 2019 and consists of 85,381 quarterly earnings announcements.

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TABLE 5
Horizon of Volatility

Panel A: Descriptive statistics for portfolios sorted by RiskInfoTerm

	Q1 (Low)	Q2	Q3	Q4	Q5 (High)	High-Low
RiskIn fo$_{30}$	−0.006	−0.001	0.000	0.002	0.009	0.015
	−(10.75)	−(5.84)	(0.35)	(4.51)	(8.73)	(13.68)
RiskIn fo$_{182}$	0.016	0.002	0.000	−0.001	−0.008	−0.024
	(8.88)	(2.97)	−(0.08)	−(2.10)	−(4.87)	−(10.23)
VolTime[	−0.498	−0.363	−0.313	−0.334	−0.380	0.118
	−(9.28)	−(8.29)	−(7.20)	−(7.47)	−(6.52)	(5.81)
VolTime$_{STND}$	−0.081	−0.007	0.025	0.021	0.012	0.092
	−(7.89)	−(0.91)	(3.64)	(3.65)	(1.14)	(6.55)

Panel B: Regression of RiskInfo of different horizons

	VolTime			*VolTime*$_{STND}$		
	(1)	(2)	(3)	(4)	(5)	(6)
RiskInfoTerm	0.036***	–	0.068***	0.028***	–	0.052***
	(4.56)	–	(4.90)	(5.26)	–	(5.21)
IVTime	–	0.010*	−0.038***	–	0.009**	−0.027***
	–	(1.77)	(−3.13)	–	(2.48)	(−3.48)
Obs	84,449	84,449	84,449	84,449	84,449	84,449
Adj. R^2	0.099	0.098	0.100	0.027	0.024	0.028
Firm FE?	Y	Y	Y	Y	Y	Y
Year FE?	Y	Y	Y	Y	Y	Y

Panel A contains time-series averages across quintiles sorted by the *RiskInfoTerm*, which captures the difference in risk information derived from 30-day and 182-day options and is defined as *RiskInfo*$_{30}$ minus *RiskInfo*$_{182}$. *RiskInfo*$_{30}$ is our proxy for the amount of information about risk contained in a firm's disclosure from 30-day options, and *RiskInfo*$_{182}$ is defined analogously for 182 options. *VolTime* is the post-announcement volatility term-structure, defined as the difference in the standard deviation of firm's daily market-adjusted returns from five to 35 trading days after the earnings announcement, relative to the standard deviation of firm's daily returns from 36 to 187 trading days after the earnings announcement, where we conservatively skip the first five trading days post-announcement to avoid any mechanical correlations because of overlap in the measurement windows.. *VolTime*$_{STND}$ standardizes the post-announcement volatility term-structure, defined as level difference in volatility after standardizing both components each calendar quarter to have a mean zero and unit standard deviation. Panel B contains regression results, where we control for *SIZE* defined as the log of market capitalization, *LBM* defined as the log of firm's book-to-market ratio, and *SURP* defined as the firms' analyst-based surprise at their earnings announcement. Firm and year fixed effects are used throughout. ΔIV_{30} is the change in 30-day implied variance from standardized options in the three-day window centered on firms' announcement date and ΔIV_{182} is defined analogously. *IVTime* is the level difference in the two implied variance change measures. The parentheses contain *t*-statistics based on 85,381 standard errors clustered by firm and year.

variables are included in multivariate tests. The two charts within figure 3 illustrate that the predictive relation of ΔIV_{30} with our outcomes of interest either flatten toward zero or take on the opposite predicted sign when studied in conjunction with *RiskInfo*$_{30}$. These findings are consistent with *RiskInfo*$_{30}$ removing some of the measurement error in ΔIV_{30} as a proxy for risk information by mitigating the influence of mechanical changes in implied variance around the announcement date.

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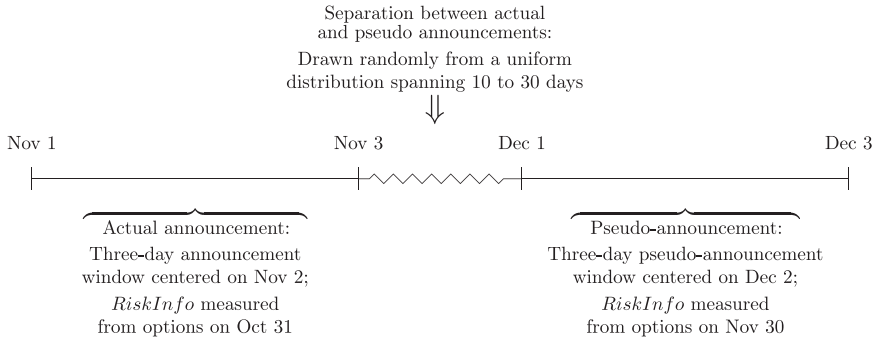
4. Extensions

4.1 PSEUDO-EARNINGS ANNOUNCEMENTS

In this section, we assess how the amount of risk information in earnings announcements compares to the amount of risk information investors learn on a typical nonannouncement day. This offers a benchmark for assessing the magnitude of risk information in earnings and how this has evolved. To do so, we estimate $RiskInfo_t$ for a sample of "pseudo" earnings announcements and compare these estimates with those from firms' actual announcements. We create pseudo-announcement dates by randomly selecting a nonannouncement date in between firms' quarterly earnings announcements.

To illustrate our approach, the graphic below contains a timeline of actual and pseudo-announcement dates used to compare quantities of risk information. The timeline uses firms with a November 2nd announcement date as an example and ignores weekends and trading holidays, to illustrate the separation between actual and pseudo-announcement dates. Actual announcement dates reflect the quarterly earnings announcement dates imputed from Compustat and I/B/E/S. We calculate pseudo-announcement dates by adding a randomly selected number of trading days to the actual announcement date. We randomly select the numbers by drawing from a uniform distribution spanning 10 to 30 days after the announcement. We draw from this window of days because we are interested in measuring risk information on nonannouncement days and hope to avoid volatility associated with future earnings announcements. Thus, we select days from this window to reduce the influence of anticipated volatility associated with firms' subsequent earnings announcements on option implied volatilities.



To help gauge the relative contribution of risk information from firms' quarterly earnings announcements versus normal dates, we compare estimates of the absolute magnitude of $RiskInfo_{30}$ when holding constant the firm and fiscal quarter. Specifically, we scale the absolute magnitude of $RiskInfo_{30}$ corresponding to firms' actual earnings announcement dates to the equivalent measure for firms' pseudo-announcement dates. In figure 4, we calculate the average ratio for each year in our sample.

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FIG 4.—Ratio of absolute RiskInfo for actual versus pseudo-announcement dates. This figure plots the time-series average ratio of absolute risk information for firms' actual versus pseudo-announcement dates. $RiskInfo_{30}$ is our proxy for the amount of information about risk contained in a firm's disclosure from 30-day options. Absolute risk information equals the absolute value of $RiskInfo_{30}$. For each actual announcement date in our main sample, we calculate risk information around a pseudo-announcement date by randomly selecting a nonannouncement date in the window of 10 to 30 days after firms' actual announcement date. For both actual and pseudo-announcement dates, we define scaled *SRiskInfo* as risk information scaled by diffusive variance implied from 30-day option prices.

Figure 4 shows that earnings announcements convey significantly more risk information than nonannouncement days. Specifically, the absolute magnitude of risk information during actual announcements, shown as blue bars, is roughly five times as large as the absolute magnitudes for pseudo-announcements. Similarly, when scaled by diffusive variance as shown in the red bars, we also find the average ratio exceeds four. These patterns suggest managers concentrate the release of information about the riskiness of their firms' cash flows during earnings announcements.

4.2 LEVERAGE EFFECTS

The literature originating with Black [1976] shows that a major determinant of a stock's risk arises from the implicit leverage embedded in equity when the firm is financed in part by debt. This "leverage effect" and its associated impact on equity risk increase when a firm's equity value decreases. Thus, news regarding a firm's expected future cash flows will also revise investors' beliefs about equity risk. Given that our measure is derived from equity options, it captures an announcement's information regarding equity risk, and thus this leverage effect. This may be seen in column 6 of table 3, which shows that $RiskInfo_{30}$ is negatively associated with earnings surprises.

In certain applications, rather than quantifying information on equity risk, one may instead be interested in information on the risk of total

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assets, that is, the value that accrues jointly to equity- and debt-holders. For example, if one wishes to evaluate the decision usefulness of regulation that focuses on the riskiness of the firm's total cash flows to all stakeholders, it may be optimal to exclude this leverage effect. In the online appendix, we develop a version of $RiskInfo_{30}$ that approximately accounts for this leverage effect.

To derive this measure, we start by applying the approach developed in section 2.3 of Bharath and Shumway [2008]. This approach translates option-implied variances, which capture the variance of equity, into implied variances of total assets.[26] We then recalculate $RiskInfo_{30}$ by replacing implied equity variances with the respective implied asset variances. Using this "leverage-adjusted" measure in the case of quarterly earnings, we find qualitatively similar results to our main inferences, suggesting the leverage effect is not the exclusive source of risk information in quarterly earnings announcements.

5. *Applications*

The final section of our paper applies $RiskInfo_t$ to three distinct research settings. Each application illustrates potential uses of $RiskInfo_t$ for future researchers while also providing platforms for discussing potential limitations when applying our measure. In doing so, these applications also provide novel conceptual insights on the transmission, measurement, and implications of risk information.

5.1 REGULATORY CHANGES IN RISK INFORMATION

In our first application, we use $RiskInfo_t$ to study the impact of the SEC's decision in 2004 to require greater use of real-time disclosures of material changes in firms' financial conditions or operations. In August 2004, Congress enacted new rules as part of the Sarbanes-Oxley Act legislating that publicly traded firms employ greater use of real-time disclosures using Form 8K filings with the SEC for material changes in their financial conditions or operations between their periodic 10K/Q filings (Noh, So, and Weber [2019]).

The 2004 regulation forced firms to disclose material contracts as they occurred, which conveyed information on firms' strategic initiatives and partnerships. These contracts were previously reported with a significant delay, often spanning multiple months. Because the regulation accelerated these disclosures, we predict that affected firms convey less risk information during their earnings announcements.

Following Noh, So, and Weber [2019], we study the impact of the 2004 regulation by measuring the extent to which a given firm filed material contracts on a delayed basis with its 10K/Qs, rather than 8Ks, prior to the

[26] We are grateful to an anonymous reviewer for suggesting this approach.

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TABLE 6

Regulatory Shocks to Risk Information at Earnings Announcements

	(1)	(2)	(3)	(4)
Post × Pre-Period Non-8K Material Contracts	−0.010**	−0.009**	−0.009**	−0.009**
	(−2.37)	(−2.13)	(−2.16)	(−2.16)
Log (Material Contracts)	0.000	0.000	0.000	0.000
	(0.37)	(0.25)	(0.19)	(0.17)
SIZE	–	−0.001***	−0.001***	−0.001***
	–	(−5.11)	(−3.12)	(−3.25)
LBM	–	–	0.002**	0.002**
	–	–	(2.41)	(2.21)
SURP	–	–	–	−0.078***
	–	–	–	(−3.73)
Obs	22,481	22,481	22,473	22,473
Adj. R^2	0.330	0.333	0.333	0.334
Firm FE?	N	Y	Y	Y
Year-Quarter FE?	N	Y	Y	Y

This table contains results from a regression of absolute risk information from 30-day options, $AbsRiskInfo_{30}$, on firms' use of material contract filings *not* filed as 8Ks prior to the 2004 SEC Regulation mandating that material contracts be filed on a more timely basis via 8Ks. For this analysis, we limit the sample to firms that have at least one Exhibit 10 *not* filed as an 8K in the pre-period. $AbsRiskInfo_{30}$ is the absolute value of our proxy for the amount of information about risk contained in a firm's disclosure from 30-day options. The sample consists of 22,481 earnings announcement observations corresponding to firm-quarters in the three years immediately before and after the regulation. *Pre-Period Non-8K Exhibit 10s* is defined as the firm's average number of material contract filings *not* filed as an 8K per pre-regulation quarter scaled by the firm's average number of 8K filings per preregulation quarter. Note that *Pre-Period Non-8K Exhibit 10s* and the indicator for *Post* are absorbed by firm-fiscal quarter fixed effects and year-quarter fixed effects, respectively, because of perfect multicollinearity. We estimate and report *t*-statistics in parentheses based on standard errors clustered by firm.

2004 regulation. The intuition for these tests is that, after the 2004 regulation, these firms were required to disclose material contracts as 8Ks and on a much more timely basis compared to their 10K/Qs. Table 6 contains estimates from the following regression of the absolute value of risk information on the preperiod incidence of material contract filings not filed as 8Ks:

$$\begin{aligned} AbsRiskInfo_{30,i,q} = {} & \alpha_1 \cdot Post_q \times Pre\,Period\,Non8K\,Exhibit\,10s_i \\ & + \alpha_2 \cdot ln(1 + Exhibit\,10s)_{i,q} \\ & + \sum \beta \cdot Controls_{i,q} + \gamma_i + \psi_q + \epsilon_{i,q}, \end{aligned} \tag{14}$$

where the i and q subscripts index the firm and quarter, respectively. Our main regression estimate of interest is the interaction term between the postregulation indicator variable, $Post$, and the firm's average number of material contract filings *not* filed as an 8K per preregulation quarter scaled by the firm's average number of 8K filings per preregulation quarter, *Pre-Period Non-8K Material Contracts*. Note that the main effects for *Pre-Period Non-8K Exhibit 10s* and *Post* when estimating equation (14) are absorbed by firm- and year-quarter fixed effects, respectively, because of perfect multicollinearity. For this analysis, we limit the sample to 22,481 earnings

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announcements in the three years immediately before and after the regulation to balance statistical power against the possibility of introducing potential confounding events.

Table 6 contains results from estimating equation (14). Consistent with our main prediction, the estimated α_1 coefficient is negative, indicating affected firms that filed more material contracts outside of 8Ks in the preperiod conveyed less risk information during their earnings announcement after the 2004 regulation. These findings suggest the regulation helped shape price discovery between earnings announcements by pulling risk information forward into real-time disclosures.[27]

The results in table 6 examine how expanded use of 8Ks affected information conveyed during anticipated information events, which our measure is particularly well-suited to capture. By contrast, if researchers' goal were to study events such as 8Ks filings that are arguably unanticipated, our measure is unlikely to offer improvement relative to the use of simple changes in implied volatilities. This is because the key innovation of our approach is that it embeds an expectation model about the anticipated arrival of information that is less likely effective in cases where investors do not anticipate an impending information release.

5.2 COMPARISON TO TEXT-BASED PROXIES

In our second application, we apply $RiskInfo_t$ to study the information content of text-driven proxies of the risk information contained in firms' annual 10K fillings with the SEC. Specifically, we compare $RiskInfo_t$ calculated around firms' 10K filing dates to the within-firm change in the number of uncertainty-related words, as defined by Loughran and McDonald [2011], contained in firms' 10Ks as part of the section labeled "Management Discussion and Analysis" (MDA) from WRDS's Readability and Sentiment file. Loughran and McDonald [2011] demonstrate that the quantity of such words relates to future return volatility, suggesting that this quantity may proxy for information on risk. We measure within-firm changes relative to firm's 10K in the prior fiscal year.

Table 7 reports averages of volatility relative to firms' 10K filing date across quintiles sorted by $RiskInfo_{30}$ in panel A, the change in the total number of uncertainty words in the 10K in panel B, and the change in fraction of uncertainty words in the 10K in panel C. Analogous to our earnings announcement tests, we report both the levels of and changes in firms' volatilities in the 30 days prior to and following firms' 10K filing date.

Panel A shows $RiskInfo_{30}$ is not significantly related to the *levels* of lagged or future volatility relative to 10K filing dates. However, we also find $RiskInfo_{30}$ again relates positively to *changes* in return volatility relative to firms' 10K filing date, consistent with investors learning about firms' riskiness via their 10Ks and incorporating the information into option prices.

[27] This is consistent with other recent work documenting that earnings risk premia have shifted forward in time from earnings to the release of 8-Ks Heitz, Narayanamoorthy, and Zekhnini [2020].

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TABLE 7

Evaluating Text-Based Measures of Risk Information

	Q1 (Low)	Q2	Q3	Q4	Q5 (High)	High-Low
Panel A: Averages by RiskInfo$_{30}$						
$RiskInfo_{30}$	−0.014	−0.002	0.000	0.003	0.020	0.034
	−(18.26)	−(11.81)	−(0.79)	(7.38)	(12.82)	(17.30)
$VLTY_{30,Pre-10K}$	0.034	0.024	0.022	0.025	0.033	−0.001
	(31.63)	(30.02)	(27.89)	(24.15)	(28.76)	−(1.46)
$VLTY_{30,Post-10K}$	0.033	0.024	0.023	0.025	0.034	0.000
	(29.07)	(29.56)	(24.71)	(23.14)	(25.23)	(0.42)
$\Delta VLTY_{30,Post-10K}$	−0.095	−0.007	0.072	−0.028	0.044	0.139
	−(1.13)	−(0.11)	(1.25)	−(0.38)	(0.50)	(2.23)
Panel B: Averages by changes in total number of uncertainty words in 10K						
$RiskInfo_{30}$	0.001	0.001	0.000	0.000	0.002	0.001
	(2.43)	(1.06)	(0.81)	−(0.82)	(2.81)	(1.25)
$VLTY_{30,Pre-10K}$	0.030	0.026	0.026	0.025	0.028	−0.002
	(30.48)	(27.62)	(23.94)	(29.06)	(28.88)	−(3.08)
$VLTY_{30,Post-10K}$	0.029	0.026	0.026	0.025	0.028	−0.001
	(26.23)	(25.35)	(22.44)	(28.03)	(24.42)	−(1.82)
$\Delta VLTY_{30,Post-10K}$	−0.036	0.005	0.001	0.010	0.024	0.060
	−(0.47)	(0.06)	(0.01)	(0.14)	(0.30)	(1.22)
Panel C: Averages by changes in fraction of uncertainty words in 10K						
$RiskInfo_{30}$	0.001	0.000	0.001	0.000	0.002	0.000
	(2.21)	(0.94)	(0.98)	(0.94)	(2.16)	(0.28)
$VLTY_{30,Pre-10K}$	0.030	0.026	0.026	0.025	0.026	−0.004
	(29.13)	(24.82)	(25.54)	(28.00)	(28.31)	−(7.27)
$VLTY_{30,Post-10K}$	0.029	0.026	0.027	0.025	0.026	−0.003
	(24.74)	(22.80)	(14.78)	(25.22)	(26.87)	−(4.78)
$\Delta VLTY_{30,Post-10K}$	−0.065	−0.014	0.193	−0.002	0.024	0.089
	−(0.81)	−(0.19)	(1.21)	−(0.02)	(0.34)	(1.81)

Panel D: Regressions of changes in volatility

	(1)	(2)	(3)	(4)	(5)	(6)
$RiskInfo_{30}$	2.501***	–	2.576***	–	2.848***	2.574***
	(3.31)	–	(2.94)	–	(3.39)	(2.95)
Δ *Uncertain Words (Level)*	–	−0.004	−0.006	–	–	–
	–	(−0.16)	(−0.21)	–	–	–
Δ *Uncertain Words (%)*	–	–	–	0.085	0.085	0.081
	–	–	–	(1.49)	(1.46)	(1.42)
DeltaIV30	–	–	25.725	–	–	25.388
	–	–	(1.19)	–	–	(1.18)
SIZE	–	–	0.095**	–	–	0.096**
	–	–	(2.54)	–	–	(2.55)
LBM	–	–	0.015	–	–	0.018
	–	–	(0.19)	–	–	(0.23)
Obs	0.298	0.311	0.314	0.312	0.313	0.315

(*Continued*)

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TABLE 7—*(Continued)*

Panel D: Regressions of changes in volatility

	(1)	(2)	(3)	(4)	(5)	(6)
Adj. R^2	22,649	18,766	18,766	18,767	18,767	18,767
Firm FE?	Y	Y	Y	Y	Y	Y
Year FE?	Y	Y	Y	Y	Y	Y

Panels A through C report averages of volatility relative to firms' 10K filing date, across quintiles sorted by $RiskInfo_{30}$, the within-firm change in fraction of uncertainty words in the 10K relative to the prior year (denoted Δ*Uncertain Words (Level)*), and the within-firm change in the total number of uncertainty words in the 10K relative to the prior year (denoted Δ*Uncertain Words (%)*). $RiskInfo_{30}$ is our proxy for the amount of information about risk contained in a firm's 10K from 30-day options. Uncertainty words come from the Wharton Research Data Services Readability and Sentiment file corresponding to firms' 10K SEC filings. Quintiles are formed on a rolling basis, using the distributional breakpoints from the prior calendar quarter. High-Low denotes the difference between the first and fifth quintile. $VLTY_{30,Pre-10K}$ is the standard deviation of firm's daily returns over the 30 trading days ending five days prior to the 10K filing date. $VLTY_{30,Post-10K}$ is the standard deviation of firms' daily returns over the 30 trading days beginning five days after the 10K filing date. $\Delta VLTY$ is the post-announcement change in return volatility, defined as the level change in $VLTY_{30,Post-10K}$ relative to $VLTY_{30,Pre-10K}$, and multiplied by 100. The parentheses in panels A through C contain t-statistics derived from the quarterly time-series average of each variable. The sample for this analysis consists of 25,245 firm-years from 1996 through 2019. Panel D contains panel regressions of $\Delta VLTY$ with firm- and year-fixed effects. The parentheses contain *t*-statistics based on standard errors clustered by firm and year.

In contrast, panels B and C show that the frequency and share of uncertainty words relate positively to the *level* of firms' past and future return volatilities, consistent with the findings of Loughran and McDonald [2011]. However, these risk-word–based proxies are either insignificantly related or weakly related to *changes* in return volatility. Panel D shows that, in panel regressions, neither text-based proxy relates significantly to changes in volatility. Panels B and C also show that text-driven proxies fail to track variation in $RiskInfo_t$. These findings suggest risk-word–based proxies are more likely to reflect managers' acknowledgment of or explanation for the *level* of firms' risks, whereas $RiskInfo_t$ captures the novel information investors glean about risks from firms' 10Ks.[28]

An important note of caution for future researchers is that our approach is ill-suited for studying the usefulness of textual disclosures for other purposes, for example, for studying managerial beliefs that have already been conveyed publicly through other media. Thus, our tests should not be interpreted as evidence that textual disclosures are devoid of content or usefulness. Instead, our tests in table 7 are designed to illustrate how our measure can be used to detect the extent of novel risk information contained in an event.

5.3 MARKET-WIDE EVENTS

In our final application, we use options on the S&P500 index to study the market-wide risk information conveyed by FOMC meetings. As a central

[28] We find similar results when using levels, rather than changes, in uncertainty words in 10Ks to measure risk information. In our online appendix, we also examine within-firm changes in uncertainty words in firms' earnings announcement conference calls and find similar evidence suggesting our findings in table 7 extend beyond 10Ks.

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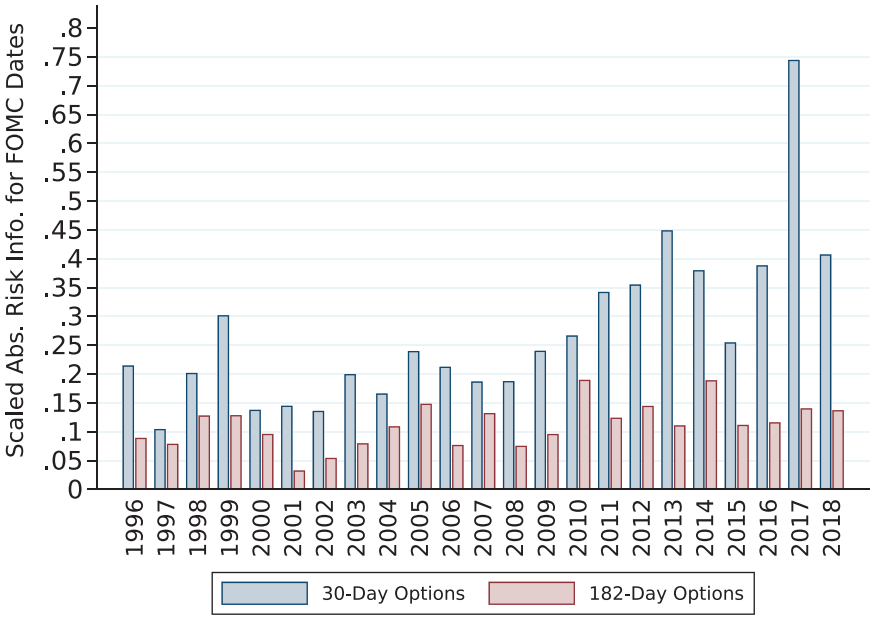

FIG 5.—Absolute RiskInfo around FOMC dates. This figure plots the time-series average of scaled absolute risk information for Federal Open Market Committee (FOMC) meeting dates. $RiskInfo_{30}$ is our proxy for the amount of information about risk contained in the FOMC announcement, which we derive from 30-day options on the S&P500 index (SPY). Absolute risk information equals the absolute value of $RiskInfo_{30}$. We measure scaled absolute risk information as the absolute value of $RiskInfo_{30}$ and $RiskInfo_{182}$ scaled by the diffusive variance implied from 30-day and 182-day option prices, respectively. The sample consists of 163 FOMC dates from 1996 through 2019.

feature in national economic and financial policy, FOMC meetings have garnered considerable attention from researchers as one of the most important anticipated information events in U.S. capital markets (e.g., Lucca and Moench [2015], Cieslak, Morse, and Vissing-Jorgensen [2019]).

Our evidence suggests FOMC meetings are a nontrivial source of market-wide risk information. Specifically, we calculate a market-wide measure of our main measure derived from S&P500 index options that we refer to as *SP500 RiskInfo*$_t$. Figure 5 shows FOMC meetings convey risk information equivalent in magnitude to approximately 21% (11%) of the implied variance in the S&P500's index return over the month (six months) following the meetings.[29]

[29] The time-series average absolute daily change in implied variance for S&P500 options is approximately 13% in our sample, suggesting that the risk information conveyed by FOMC announcements is significantly greater than the amount investors' beliefs about risk change on the typical day. However, the economic importance of this figure depends both on the relevance of market volatility to a particular researcher and the level of volatility expected over the month following FOMC meetings.

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TABLE 8

FOMC Meetings and Investor Sentiment

	(1)	(2)	(3)	(4)	(5)
SP500 RiskInfo$_{30}$	−0.313**	−0.307**	−0.303***	−0.308***	−0.324***
	(−2.51)	(−2.45)	(−3.57)	(−3.54)	(−3.52)
ΔIV$_{30}$	–	–	0.230***	0.245***	0.193**
	–	–	(3.78)	(3.52)	(2.49)
Δ *Yield*	–	–	–	0.095	0.097
	–	–	–	(1.52)	(1.52)
S&P500 Return	–	–	–	–	−0.076
	–	–	–	–	(−1.18)
N	163	163	163	162	162
Adj. R^2	0.092	0.191	0.234	0.235	0.232
Year FE?	N	Y	Y	Y	Y
Month-of-Year FE?	N	Y	Y	Y	Y

This table contains results from a regression of changes in market-wide investor sentiment on risk information for Federal Open Market Committee (FOMC) meeting dates. We measure changes in investor sentiment using the index from Baker and Wurgler [2006] in the months surrounding FOMC dates. *SP500 RiskInfo*$_{30}$ is our proxy for the amount of information about risk contained in the FOMC announcement, which we derive from 30-day options on the S&P500 index (SPY) directly surrounding the FOMC date. ΔIV_{30} is the change in 30-day implied variance on the S&P500 index from standardized options in the three-day window centered on the FOMC date. Δ*Yield* is the change in 10-year Treasury yield in the three-day window surrounding the FOMC date. *S&P500 Return* is the return on the S&P500 index (SPY) for the three-day window surrounding the FOMC date. We estimate and report t-statistics in parentheses based clustered robust standard errors, clustered by year. The sample consists of 163 FOMC dates from 1996 through 2019.

In table 8, we examine the link between risk information conveyed by FOMC meetings and changes in investor sentiment. Specifically, we run regressions of changes in the investor sentiment index from Baker and Wurgler [2006] by measuring the index at the month-ends directly surrounding the FOMC meetings. To illustrate that *SP500 RiskInfo*$_t$ captures information not summarized by common proxies of the news in these meetings, these tests also include controls for changes in implied volatilities for the S&P500, changes in 10-year Treasury yield, denoted Δ*Yield*, and the return on the S&P500 index (SPY) for the three-day window surrounding the FOMC meeting date. For ease of interpretation, all variables are standardized to have a zero mean and unit standard deviation.

The results in table 8 highlight a robust negative link between *SP500 RiskInfo*$_t$ and changes in investor sentiment. These findings are consistent with investor sentiment falling when investors revise upward their expectations of risk in response to FOMC meetings. In economic terms, the coefficients hover around 0.3 in magnitude indicating a one standard deviation increase in *SP500 RiskInfo*$_t$ corresponds to roughly to a one-third standard deviation decrease in investor sentiment.[30] The table also shows that ΔIV is *positively* associated with sentiment, which highlights the importance

[30] Because *SP500 RiskInfo*$_t$ equals the difference between realized and expected volatilities, the positive coefficient in ΔIV_{30} suggests that sentiment declines following FOMC announcements with expected risk news. Although not the focus of our study, this finding is consistent

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of correcting for $AnnVar$ in this setting. This indicates that expected announcement volatility, as captured by $AnnVar$, is negatively associated with sentiment. Together with the negative link between *SP500 RiskInfo*$_t$ and sentiment, this implies that sentiment is negatively linked to risk perceptions both on and after FOMC announcement dates.

The results in table 8 are subject to the caveat that the VRP may change around macroeconomic news events, such as FOMC meetings, which, as discussed in section 2.2, may lead to a bias in the measure. As previously discussed, the extent of the bias in the measure can be gauged using the sample average of *RiskInfo*$_t$. Here, we find that the mean levels of *SP500 RiskInfo*$_{30}$ and *SP500 RiskInfo*$_{182}$ equal -0.00026 and -0.00034, respectively. To put this in perspective, this equals 26% and 12% of average absolute level of risk information revealed by the event over 30 days and 182 days, respectively—suggesting a material degree of bias. In forecasting applications, whether this presents a concern depends upon the extent to which the amount of uncertainty these meetings resolve varies over time. For example, if macroeconomic events have a homogeneous impact on uncertainty over the future return variance, they will reduce the VRP by a similar amount, leading to a constant downward bias in our measure.

By contrast, the negative link between *SP500 RiskInfo*$_{30}$ and changes in sentiment we document in table 8 may be overstated to the extent sentiment systematically rises for FOMC meetings that command higher VRPs. Although we are not aware of a reason to anticipate this form of bias, we emphasize that the appropriateness of *SP500 RiskInfo*$_t$ for studying macroeconomic events is context dependent. Note applying our measure at the market level is analogous to the numerous forecasting applications of model-free implied moments found in the literature, which also depend both upon forward-looking return moments and their embedded risk premia (Christoffersen, Jacobs, and Chang [2013]).

Taken together, the results in this section provide novel insights on the implications of disclosure regulation, the efficacy of textual-based measures, and the link between risk information and investor sentiment. In doing so, these tests help illustrate that our measure enables a wide array of research on the multidimensional nature of information conveyed to market participants via anticipated announcements.

6. Conclusion

We develop a measure of how information events impact investors' expectations of risk that is broadly applicable and simple to implement. We derive this measure from an option-pricing model, where investors anticipate an announcement that simultaneously conveys information on the announcer's future cash flows and risk profile. Our model also shows researchers cannot estimate risk information from equity prices alone.

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with option prices anticipating higher volatility associated FOMC dates that convey more negative information.

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We empirically implement the measure using firms' earnings announcements and find results that closely align with our model's predictions. In doing so, we highlight methodological implications for researchers using simple changes in option-implied volatilities to study risk information gleaned from information events. We also apply our measure to study disclosure regulation, the efficacy of text-based proxies, and market-wide events, providing novel evidence on the transmission, measurement, and implications of risk information.

Given that investors' risk perceptions play a central role in numerous research settings, our measure has a number of potential applications. For example, applied as a dependent variable, $RiskInfo_t$ may be used to study the efficacy of risk-based disclosure requirements, to gauge whether changing audit and verification standards impact investors' perceptions of the risk conveyed through earnings announcements, or to assess the plausibility of risk-based explanations for announcement-based return predictors by examining how they correlate with risk information. The measures of the *quantity* of risk information provided by an event defined in table 1, which utilize the absolute value of $RiskInfo_t$, may be especially useful in these contexts.

In addition, applied as an independent variable, our measure may be used to study the economic consequences of shocks to investors' beliefs about risk. For example, $RiskInfo_t$ can be applied to study how sophisticated versus unsophisticated investors change their portfolio holdings in response to risk information, the usefulness of risk information in debt contracting and optimal compensation design, and the role of risk information in spurring coverage from information intermediaries, such as analysts, rating agencies, and the financial press. As is the case when applying other metrics of information quality, one must be cautious in making causal inferences when applying the measure as an independent variable and consider the use of quasi-experimental methods (Leuz and Wysocki [2016]). Differences in the amount of information firms provide on their risks likely relate to other firm characteristics, many of which may be unobservable.

APPENDIX A

An Equilibrium Model

In this appendix, we develop an equilibrium model in which the firm's stock price endogenously follows the process discussed in the main text. Specifically, we consider a model in which investors face uncertainty regarding both the covariance between a firm's future cash flows and consumption and idiosyncratic cash-flow volatility. On an anticipated date, news arrives regarding the level of future cash flows, their variance, and their covariance with consumption.

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Formally, consider a representative investor who trades in a firm's stock and options over a time interval $[0, \infty)$. This investor has time-additive CRRA utility:

$$U_t(\{\tilde{c}_s\}_{s\in[t,\infty)}) = \int_t^\infty \frac{e^{-\delta(s-t)}\tilde{c}_s^{1-\alpha}}{1-\alpha} ds,$$

and has consumption that evolves as[31]:

$$\frac{d\tilde{c}_t}{\tilde{c}_t} = \sum_{i=1}^{k-1} \sigma_{f,i} dW_{i,t}^c.$$

The Brownian motions $\{dW_{i,t}^c\}_{i\in\{1,\ldots,k-1\}}$ denote the priced risk factors in the economy, and the coefficients $\sigma_{f,i}$ determine their associated risk premia. On date T the firm pays a dividend $\tilde{x}_T$ that represents its future value.[32] Assume that investors observe $\tilde{x}_t$ on date t, where $\tilde{x}_t$ is a stochastic process that converges toward $\tilde{x}_T$. Specifically, we assume that $\tilde{x}_t$ follows a Heston-type process:

$$d\tilde{x}_t = \tilde{x}_t \mu_x dt + \tilde{x}_t \sum_{i=1}^{k-1} \tilde{\beta}_i dW_{i,t}^c + \tilde{x}_t \tilde{\sigma}_{\varepsilon,t} dW_t^\varepsilon + dN_t^x;$$

$$d\tilde{\sigma}_{\varepsilon,t}^2 = \kappa_\varepsilon\left(\tilde{\sigma}_{\varepsilon,t}^2 - \theta_\varepsilon\right)dt + \xi_\varepsilon \tilde{\sigma}_{\varepsilon,t} dW_t^{\sigma\varepsilon},$$

where all Brownian motions in the model are independent. The investor observes W_t^ε at date t, which leads the equilibrium price to continuously evolve. Note the terms $\tilde{\beta}_i$, which are independent of all other random variables, capture the firm's systematic risk-factor exposures, whereas $\tilde{\sigma}_{\varepsilon,t}$, which follows a CIR process, captures the firm's time-varying idiosyncratic volatility. Although we could allow for alternative stochastic processes for investor learning, we will see that this process causes the firm's price to follow a Heston process in equilibrium, as commonly used in the empirical option-pricing literature.

Again, at time τ_D, which is known in advance to the investor, an announcement occurs. As a simple way to incorporate information in this announcement that influences expected future cash flows, we assume the firm's dividend process exhibits a jump on the date of the information event, which is captured by dN_t^x, where:

$$N_t^x = \tilde{x}_{\tau_D^-}(\exp(\tilde{J}_x) - 1) * \mathbf{1}(t \geq \tau_D),$$

and where $\tilde{J}_x$ is independent of other variables in the model. To capture the risk information in the event in a parsimonious fashion, we assume that

[31] Growth in consumption can be incorporated into the model with no change in the results, outside of a shift in the risk-free rate.

[32] We could instead assume a series of periodic dividends from period T onward, with no material change in the results. Intuitively, we may view $\tilde{x}_T$ as the discounted present value of all of the firm's future dividends.

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this announcement perfectly resolves all distributional uncertainty, that is, it perfectly reveals $\{\tilde{\beta}_i\}_{i\in\{1,\ldots,k-1\}}$ and $\{\tilde{\sigma}_{\varepsilon,t}\}_{t\in[\tau_D,T]}$.

We next derive the firm's equilibrium price process.[33]

Proposition A.1. *Let* $r = \delta - \frac{\alpha(1+\alpha)}{2}\sum_{i=1}^{k-1}\sigma_{f,i}^2$. *The firm's stock price just prior to the announcement,* $\lim_{t\to\tau_D^-} P_t$, *satisfies:*

$$\lim_{t\to\tau_D^-} P_t = \tilde{x}_{\tau_D^-}\mathbb{E}_{\tau_D^-}\left[\exp\left((T-t)\left(\mu_x - r - \alpha\sum_{i=1}^{k-1}\tilde{\beta}_i\sigma_{f,i}\right) + \tilde{J}_x\right)\right], \tag{A.1}$$

and the firm's price at a date following the announcement, $t \geq \tau_D$ *satisfies:*

$$P_t = \tilde{x}_t \exp\left((T-t)\left(\mu_x - r - \alpha\sum_{i=1}^{k-1}\tilde{\beta}_i\sigma_{f,i}\right)\right). \tag{A.2}$$

On date τ_D, *the firm's price jumps, such that:*

$$\frac{P_t}{\lim_{t\to\tau_D^-} P_t} = \frac{\exp\left(-\alpha(T-t)\sum_{i=1}^{k-1}\tilde{\beta}_i\sigma_{f,i} + \tilde{J}_x\right)}{\mathbb{E}_{\tau_D^-}\left[\exp\left(-\alpha(T-t)\sum_{i=1}^{k-1}\tilde{\beta}_i\sigma_{f,i} + \tilde{J}_x\right)\right]}.$$

Following date τ_D, *the firm's value evolves as:*

$$\frac{dP_t}{P_t} = \left(r + \alpha\sum_{i=1}^{k-1}\tilde{\beta}_i\sigma_{f,i}\right)dt + \sum_{i=1}^{k-1}\tilde{\beta}_i dW_{i,t}^c + \tilde{\sigma}_{\varepsilon,t}dW_t^{\varepsilon}. \tag{A.3}$$

$$\textit{where } d\tilde{\sigma}_{\varepsilon,t}^2 = \kappa_\varepsilon\left(\tilde{\sigma}_{\varepsilon,t}^2 - \theta_\varepsilon\right)dt + \xi_\varepsilon\tilde{\sigma}_{\varepsilon,t}dW_t^{\sigma\varepsilon}. \tag{A.4}$$

Proof. Let $\tilde{C}_t \equiv \ln(\tilde{c}_t)$. Standard arguments imply that, in this economy, a necessary condition for the representative agent's first-order condition to be satisfied is that there exists a stochastic-discount factor process M_t, where (e.g., Back [2010]):

$$\begin{aligned} M_t &= e^{-\delta t}\frac{\tilde{c}_t^{-\alpha}}{c_0^{-\alpha}} \\ &= \exp\left[-\delta t - \alpha\left(\tilde{C}_t - C_0\right)\right]. \end{aligned} \tag{A.5}$$

This implies the date t price of a bond paying off 1 dollar at date $s > t$ equals:

$$\mathbb{E}_t\left[\frac{M_s}{M_t}\right] = \mathbb{E}_t[\exp[-\delta(s-t) - \alpha(\tilde{C}_s - C_t)]]. \tag{A.6}$$

[33] Note that, prior to date τ_D, investors can use the correlation between $\tilde{x}_t$ and the respective risk factors to update their beliefs regarding the firm's risk-factor exposures. If $\tilde{\sigma}_{\varepsilon,t} \to 0$ and the individual consumption factors were observable, this updating would be perfect, as investors could simply regress the firm's variance on the factor variances to arrive at the respective betas. However, the presence of nontrivial idiosyncratic risk $\tilde{\sigma}_{\varepsilon,t}$ leads to imperfect updating, such that investors face residual uncertainty regarding the factor exposures leading up to the announcement date.

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Note that, as of date t,

$$-\alpha(\tilde{C}_s - \tilde{C}_t) \sim N\left(\frac{\alpha(s-t)\sum_{i=1}^{k-1}\sigma_{f,i}^2}{2}, \alpha^2(s-t)\sum_{i=1}^{k-1}\sigma_{f,i}^2\right). \tag{A.7}$$

Therefore, $\mathbb{E}_t[\frac{M_s}{M_t}] = \exp[(-\delta + \frac{\alpha(1+\alpha)}{2}\sum_{i=1}^{k-1}\sigma_{f,i}^2)(s-t)]$, which implies that the equilibrium instantaneous return on the bond is $r \equiv \delta - \frac{\alpha(1+\alpha)}{2}\sum_{i=1}^{k-1}\sigma_{f,i}^2$. Moreover, the stock price satisfies:

$$\begin{aligned} P_t &= \mathbb{E}_t\left[\tilde{x}_T\frac{M_T}{M_t}\right] \\ &= e^{-\delta(T-t)}\mathbb{E}_t[\tilde{x}_T\exp[-\alpha(\tilde{C}_T - \tilde{C}_t)]]. \end{aligned} \tag{A.8}$$

Now, note that:

$$\begin{aligned} \tilde{x}_T &= \tilde{x}_t * \exp(\mathbf{1}(\tau_D \in (t,T])\tilde{J}_x) \\ &\quad * \exp\left(\mu_x(T-t) - \frac{\int_t^T \tilde{\sigma}_{\varepsilon,y}^2 dy}{2} - \frac{(T-t)\sum_{i=1}^{k-1}\tilde{\beta}_i^2}{2} + \sum_{i=1}^{k-1}\int_t^T \tilde{\beta}_i dW_{i,y}^c + \int_t^T \tilde{\sigma}_{\varepsilon,y} dW_y^\varepsilon\right). \end{aligned} \tag{A.9}$$

Substituting, applying iterated expectations, and simplifying, we have that:

$$\begin{aligned} &\mathbb{E}_t\{\tilde{x}_T\exp[-\alpha(\tilde{C}_T - C_t)]\} \\ &= \mathbb{E}_t\left\{\mathbb{E}_t\left[\tilde{x}_T\exp[-\alpha(\tilde{C}_T - C_t)]\middle|\{\tilde{\beta}_i\}_{i=1}^{k-1}, \{\tilde{\sigma}_{\varepsilon,y}\}_{y\in[t,T]}\right]\right\} \\ &= \exp(\alpha\tilde{C}_t + \mu_x(T-t))\tilde{x}_t\mathbb{E}_t\left\{\exp\left(\mathbf{1}(\tau_D \in (t,T])\tilde{J}_x - \frac{\int_t^T \tilde{\sigma}_{\varepsilon,y}^2 dy}{2} - \frac{(T-t)\sum_{i=1}^{k-1}\tilde{\beta}_i^2}{2}\right)\right. \\ &\quad \left. * \mathbb{E}_t\left[\exp\left(\sum_{i=1}^{k-1}\int_t^T \tilde{\beta}_i dW_{i,y}^c - \alpha\tilde{C}_T + \int_t^T \tilde{\sigma}_{\varepsilon,y} dW_y^\varepsilon\right)\middle|\{\tilde{\beta}_i\}_{i=1}^{k-1}, \{\tilde{\sigma}_{\varepsilon,y}\}_{y\in[t,T]}\right]\right\}. \end{aligned} \tag{A.10}$$

To evaluate the expectations in this expression, note that, as of date t, conditional on $\{\tilde{\beta}_i\}_{i=1}^{k-1}$ and $\{\tilde{\sigma}_{\varepsilon,y}\}_{y\in[t,T]}$,

$$\sum_{i=1}^{k-1}\int_t^T \tilde{\beta}_i dW_{i,y}^c - \alpha\tilde{C}_T \sim N\left(-\alpha\tilde{C}_t + \frac{\alpha(T-t)\sum_{i=1}^{k-1}\sigma_{f,i}^2}{2}, (T-t)\sum_{i=1}^{k-1}(\tilde{\beta}_i - \alpha\sigma_{f,i})^2\right).$$

Furthermore, $\int_t^T \tilde{\sigma}_{\varepsilon,y} dW_y^\varepsilon | \{\tilde{\sigma}_{\varepsilon,y}\}_{y\in[t,T]} \sim N(0, \int_t^T \tilde{\sigma}_{\varepsilon,y}^2 dy)$. Thus, expression (A.8) reduces to:

$$\begin{aligned} P_t &= \tilde{x}_t\mathbb{E}_t\left[\exp\left((\mu_x - \delta)(T-t) + (T-t)\sum_{i=1}^{k-1}\frac{\left[(\tilde{\beta}_i - \alpha\sigma_{f,i})^2 + \alpha\sigma_{f,i}^2 - \tilde{\beta}_i^2\right]}{2}\right.\right. \\ &\quad \left.\left. + \mathbf{1}(\tau_D \in (t,T])\tilde{J}_x\right)\right] \\ &= \tilde{x}_t\mathbb{E}_t\left[\exp\left(\mu_x(T-t) + (T-t)\left(-\delta + \frac{\alpha(1+\alpha)}{2}\sigma_{f,i}^2 - \alpha\sum_{i=1}^{k-1}\tilde{\beta}_i\sigma_{f,i}\right)\right.\right. \\ &\quad \left.\left. + \mathbf{1}(\tau_D \in (t,T])\tilde{J}_x\right)\right] \\ &= \tilde{x}_t\mathbb{E}_t\left[\exp\left((T-t)\left(\mu_x - r - \alpha\sum_{i=1}^{k-1}\tilde{\beta}_i\sigma_{f,i}\right) + \mathbf{1}(\tau_D \in (t,T])\tilde{J}_x\right)\right]. \end{aligned} \tag{A.11}$$

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This implies that:

$$\lim_{t\to\tau_D^-} P_t = \tilde{x}_{\tau_D^-}\mathbb{E}_{\tau_D^-}\left[\exp\left((T-t)\left(\mu_x - r - \alpha\sum_{i=1}^{k-1}\tilde{\beta}_i\sigma_{f,i}\right) + \tilde{J}_x\right)\right],$$

which verifies equality (A.1). Next, for $t \geq \tau_D$, note that:

$$\begin{aligned} P_t &= \tilde{x}_t\mathbb{E}_t\left[\exp\left((\mu_x - r)(T-t) - \alpha(T-t)\sum_{i=1}^{k-1}\tilde{\beta}_i\sigma_{f,i}\right)\right] \\ &= \tilde{x}_t\exp\left((T-t)\left(\mu_x - r - \alpha\sum_{i=1}^{k-1}\tilde{\beta}_i\sigma_{f,i}\right)\right), \end{aligned} \tag{A.12}$$

which verifies equality (A.2). Next, for $t \geq \tau_D$, let $R(t) \equiv \frac{P_t}{x_t} = \exp((T-t)(\mu_x - r - \alpha\sum_{i=1}^{k-1}\tilde{\beta}_i\sigma_{f,i}))$. Applying Ito's lemma to the above expression, for $t \geq \tau_D$,

$$\begin{aligned} dP_t &= x_t R' dt + R dx_t \\ &= x_t R\left[-\left(\mu_x - r - \alpha\sum_{i=1}^{k-1}\tilde{\beta}_i\sigma_{f,i}\right)dt + \frac{dx_t}{x_t}\right]. \end{aligned} \tag{A.13}$$

Thus,

$$\begin{aligned} \frac{dP_t}{P_t} &= -\left(\mu_x - r - \alpha\sum_{i=1}^{k-1}\tilde{\beta}_i\sigma_{f,i}\right)dt + \frac{dx_t}{x_t} \\ &= \left(r + \alpha\sum_{i=1}^{k-1}\tilde{\beta}_i\sigma_{f,i}\right)dt + \sum_{i=1}^{k-1}\tilde{\beta}_i dW_{i,t}^c + \tilde{\sigma}_{\varepsilon,t}dW_t^{\varepsilon}, \end{aligned}$$

which completes the derivation of equation (A.3). □

Proposition A.1 demonstrates the mapping between this equilibrium model and the statistical model considered in the main text. Specifically, the price process in the model matches the one in the main text upon setting $\mu_t = \mu_x$ and[34]:

$$\begin{aligned} &\tilde{\sigma}_{i,t} = \tilde{\beta}_i \text{ for } t \in [\tau_D, T],\ i = 1, \ldots, k-1; \\ &\tilde{\sigma}_{k,t} = \tilde{\sigma}_{\varepsilon,t} \text{ for } t \in [\tau_D, T]; \end{aligned} \tag{A.14}$$

$$\begin{aligned} g\big(\tilde{y}, \{\tilde{\sigma}_{i,t}\}_{i\in\{1,\ldots,k\}}\big) &= \tilde{J}_x - \ln\mathbb{E}_{\tau_D^-}[\exp(\tilde{J}_x)] - \alpha(T-t)\sum_{i=1}^{k-1}\tilde{\beta}_i\sigma_{f,i} \\ &\quad - \ln\left\{\mathbb{E}_{\tau_D^-}\left[\exp\left(-\alpha(T-t)\sum_{i=1}^{k-1}\tilde{\beta}_i\sigma_{f,i}\right)\right]\right\}. \end{aligned}$$

[34] Note the finite horizon in this model is unimportant, as the terminal dividend date T may be arbitrarily far into the future.

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Although this process matches the one applied in the measure's derivation, it violates two of the assumptions used in the Black-Scholes model, which may lead to error when applying the measure using Black-Scholes implied variances. First, the price process exhibits stochastic volatility (a feature that is necessary in order to incorporate risk information). Second, the price drift includes the random term $\alpha \sum_{i=1}^{k-1} \tilde{\beta}_i \sigma_{f,i}$, which is negatively correlated with the jump in price on the announcement date. That is, the firm's price exhibits negative autocorrelation. Intuitively, if the announcement suggests that the firm's betas are high, its risk premium will rise, causing the price to drop. As time passes and information arrives, uncertainty falls and this increase in the risk premium dissipates. Relative to the case in which there is no such correlation, this effect reduces the amount of uncertainty investors face about the firm's future price, which can cause $AnnVar$ to overshoot the actual return variance anticipated on the announcement date. We show in the next section that for reasonable parameter ranges, the resulting error in the measure is small.

APPENDIX B

Simulations

In this section, we simulate the model to assess the performance of the Black-Scholes approximation used to derive our measure. Note in order to simulate the model, it is necessary to assume a specific distribution for $\tilde{y}$ and a functional form for $g(\cdot)$, as well as specific processes followed by firms' risk exposures. We base the simulations on the distributions and functional forms in the equilibrium model documented in appendix A. Note this model allows for general distributions of $\tilde{J}_x$ and $\{\tilde{\beta}_i\}_{i\in\{1,\ldots,k-1\}}$. We assume that $\tilde{J}_x \sim N(-\frac{1}{2}\sigma_J^2, \sigma_J^2)$ and adjust θ_ε to match $\sigma_{\varepsilon,0}^2$ in order to ensure that the stock and variance price paths begin in steady state. We focus on a single risk factor $W_{1,t}^c$, and thus a single risk-factor exposure, $\tilde{\beta}_1$ and assume that it is uniformly distributed on $[\beta_L, \beta_H]$.

In conducting the simulations, we first simulate stock price, option price, and variance paths, and calculate $RiskInfo_t$ based on the simulated stock and option-price paths. Consistent with our empirical analyses, we calculate the measures using ATM option prices and examine 30-day and 182-day versions of our measure. To assess the measure's performance, we then calculate the average absolute deviation between the measure and the true change in investors' beliefs regarding the variance of stock returns as a fraction of the average level of stock-return variance over the relevant horizon.

Recall in the model in appendix A that the firm's idiosyncratic return variance follows a CIR process. We conduct the simulations over a 6^3 grid of the parameters for this process, σ_J^2, ξ_ε, and $\theta_\varepsilon = \sigma_{\varepsilon,0}^2$. We choose to examine the impact of ξ_ε on the performance of the measure as this parameter controls the extent to which the volatility process exhibits time-variation and thus the extent to which the model deviates from the Black-Scholes

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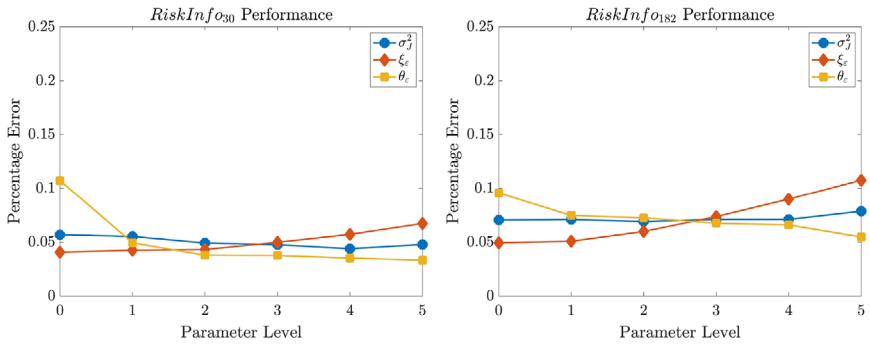

FIG B.1.—This figure displays the results of our simulations. In these simulations, we vary σ_f^2, ξ_ε, and $\tilde{\sigma}_{\varepsilon,0}^2 = \theta_\varepsilon$ on a 6^3 dimensional grid ranging uniformly over $[0, 0.3] \times [0, 0.3] \times [0.3, 1]$ and fix $\kappa = 2$. We then set δ such that the risk-free rate r equals 4% and choose α, $\sigma_{f,1}$, β_H and β_L such that the stock's average discount rate in excess of r is 5%, and such that, following the announcement, the discount rate changes uniformly between +20 and -20 bps. For each set of parameters, we first generate variance paths and option prices. We then calculate our measure using these option prices and calculate the percentage error as the difference between the measure and the true change in the market's beliefs regarding the firm's risk as a fraction of the average level of firm risk over the relevant horizon. The plots depict the average percentage error as the parameters vary.

model. Furthermore, we choose to examine the impact of the parameters σ_f^2 and $\theta_\varepsilon = \sigma_{\varepsilon,0}^2$ on the performance of the measure as they control the relative amount of variation in returns driven by the announcement versus diffusive sources of information. This enables us to verify that the measure's efficacy is not sensitive to the overall importance of the announcement in driving the firm's price.

We set δ to ensure that the risk-free rate r equals 4%. We choose α, $\sigma_{f,1}$, β_H, and β_L such that the stock's average discount rate in excess of r is 5%, and such that, following the announcement, the discount rate changes by an amount that varies uniformly between +20 and −20 bps, which is roughly consistent with the highest and lowest decile that we find in our sample of quarterly earnings announcements. In the online appendix, we demonstrate that, although the error rises upon allowing for more variation in the discount rate (which corresponds to a greater magnitude of information on risk-factor exposures), it remains under 10% even when allowing for 10 times this variation. We assume that the dividend is paid after the options considered expire (i.e., six months following the event). We further show in this appendix that our results are also robust to a variant of the model in which the firm is in steady state following the announcement.

Figure B.1 displays the results of the simulations. The figure shows that errors for *RiskInfo*$_{30}$ (*RiskInfo*$_{182}$) lie in the range of 3–11% (5–11%). The most significant determinant of the error in the measure is ξ_ε, which determines the extent of uncertainty regarding future idiosyncratic volatility. The overall level of idiosyncratic volatility relative to the significance of the

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announcement, as captured by σ_J^2 and θ_ε, plays a minimal role in the measure's performance.

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